Università degli Studi di Firenze
Department of Information Engineering

Master’s Degree Thesis in Artificial Intelligence

A platform for supporting onshore wind farm siting in Australia

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Contents

Introduction

Wind farm site selection is a crucial yet highly complex challenge. Optimal siting involves much more than simply identifying windy locations; it requires balancing multiple, often competing factors including, but not limited to, proximity to the electricity grid, consistency of wind resources, land use constraints, environmental concerns, community acceptance, and economic viability [3, 18, 24]. The goal is to select locations that will not only generate the most reliable energy over time but also integrate well into the broader power system and surrounding landscape. However, this decision-making process is often not systematically optimized.

This issue has become even more pressing in the context of growing climate change awareness and the global transition toward sustainable energy systems. In Australia, policy commitments such as the Victorian Renewable Energy Targets, legislated in the *Renewable Energy Act (2017)* [46], reinforce this shift. These targets include:

- 40% of Victoria's electricity generation to be renewable by 2025
- 65% by 2030
- 95% by 2035

Meeting such goals will demand not only increased renewable energy capacity, but also smarter, more coordinated deployment strategies. This underscores the need for data-driven and holistic planning tools that can guide decision-making.

Currently, wind farm development in Australia is largely driven by a bottom-up planning approach. While site selection occurs within a defined legal framework and requires approval from energy market regulators, final decisions are often shaped by the interests of local stakeholders, like landowners, and by wind resource assessments conducted over relatively short timescales. Given that wind farms typically operate for 20 to 30 years, this approach can result in long-term inefficiencies [53]. Studies have shown that such a fragmented planning process can lead to suboptimal outcomes in terms of both total energy generation and supply stability. In contrast, a top-down planning model, where wind farm locations are determined with long-term demand patterns in mind, has demonstrated the potential for significantly greater efficiency and reliability [21].

Literature Review

This literature review aims to explore existing methods for wind farm site selection, both globally and within Australia. It will examine traditional approaches as well as recent techniques that incorporate artificial intelligence.

Findings from the review will inform the design of a user-friendly, explainable decision-support platform intended for a diverse range of users. These can include both technical actors such as infrastructure planners, power operators (e.g., AEMO, VicGrid) and decision makers, as well as ordinary citizens and local communities. In this context, clear and interactive representations of site selection criteria can help build trust, reduce misunderstandings, and support more inclusive planning. This is particularly important because the question of where to produce renewable electricity is often controversial, and the acceptance of new energy infrastructures, especially by the communities living near proposed sites, is critical, and it can lead to tension between stakeholders with different priorities or levels of technical understanding. [18, 32, 45, 48].

Figure 2.1: Recent protests against wind farms in Australia. **(Center)** A protest in Wollongong against plans for an offshore wind farm, May 2025 (Photo: Anna Warr, The Advocate). **(Right)** A rally on the front lawns of Parliament House in Canberra, February 24 (Photo: Mike Bowers, The Guardian). **(Left)** A sign photographed by the author in Warrnambool, March 2025

The review includes comparisons between approaches used across different countries. The state of the art summary then identifies gaps in the existing literature, particularly the lack of explainable and accessible visual tools.

2.1 Data Inputs for Wind Farm Site Suitability Modeling

Before delving into the various methods used for wind farm site selection, it is important to first understand the types of data commonly employed in these studies. The selection of suitable wind farm sites is inherently data-driven, relying on a wide range of datasets to evaluate the feasibility, efficiency, and sustainability of potential locations. These datasets, which span technical, environmental, and social factors, are fundamental in determining the most appropriate sites for wind farm development. After reviewing numerous studies it becomes evident that the types of data used are similar, regardless of geographical location or the specific approach employed. The most common siting factors identified across these studies are summarized in the table below:

Siting Factor
Wind Speed
Distance to Protected or Wildlife Areas
Slope
Distance to Urban Centers
Distance to Roads
Distance to Transmission Lines or Substations
Life Cycle Costs
Distance to Airports
Distance to Water Bodies
Elevation
Distance to Animal Habitats or Migration Zones
Wind Power Density
Distance to Agricultural Areas
Flood Risk
Landslide Risk
Distance to Historic Places
Distance to Railroads
Distance to Military Zones

Table 2.1: Common siting factors used in (onshore) wind farm site selection

However, even with these shared factors, a significant challenge persists: the lack of standardization in how data is represented and utilized. As highlighted in [55], a consistent and standardized approach to data representation is absent across the studies. This variability in data handling can create discrepancies in the results and undermines the reliability of findings across different studies, especially if applied to the same region of interest. The wind speed, for example, may be categorized either as an economic factor [19, 36], or as a techincal factor [1, 17]; as a constraint [20] (*i.e.* setting a minimum threshold) or as an evaluation criterion [33] (taken into account in order to maximize potential energy output). Furthermore, the choice of data sources often remains unclear, with many studies failing to specify where their datasets were obtained [1, 3, 8, 9, 25, 26], which hinders transparency and replicability in site selection research.

2.2 Methods

The methods used for wind farm site selection (WiFSS) have grown more sophisticated over the past decade, driven by the growing demand for renewable energy. In general, these methods can be grouped into two main categories:

- Multi Criteria Decision Making **(MCDM)** methods
- **Non-MCDM** methods

Each group including a range of different techniques, differing in how they handle and process data.

2.2.1 Multi Criteria Decision Making (MCDM) methods

MCDM has been an active area of research since the 1970s [54] and its methods are applied across a wide range of research fields, including mathematics, energy systems, computer science, and economics [49]. MCDM methods addresses decision problems involving multiple, often conflicting criteria, typically measured in incomparable units. MCDM methods are designed to support the evaluation and selection of alternatives in complex scenarios where trade-offs are inevitable [35], such as in wind farm site selection. Determining importance coefficients (weights) for each factor of the problem is the central challenge in MCDM methods.

Analytic Hierarchy Process (AHP) In the context of wind farm site selection, the Analytic Hierarchy Process is probably the most extensively applied MCDM method, with numerous studies employing it across different regions in England [52], Saudi Arabia [11], Ecuador [51], India [43], Spain [42], among others. AHP is a rank-order weighting method, which means that each factor is considered to have a different level of importance when determining the best location for a wind farm. This method is composed of two key stages: pairwise comparisons and weighted linear combination (WLC). The pairwise comparison stage is needed to assign relative importance weights to the factors of the problem (e.g. distance from electrical grid, averaged wind speed). While WLC is used in the final step to evaluate and rank each potential wind farm location based on the calculated weights [44]. For instance, one wind farm location may have a certain distance from a national park, a specific distance from the electrical grid, and a particular average wind speed. Another location, in turn, might have different values for these same factors. Experts, using their knowledge and judgment, compare the criteria by performing pairwise scenario comparisons. Their opinions are then used to generate a comparison matrix, which is subsequently used to establish the relative weights. These weights determine which factors are more or less important in the context of the overall goal of identifying the best location. The main advantage of this method lies in its ability to break down a large problem into smaller, more manageable subproblems (such as comparing pairs of scenarios at different hierarchical levels) [2]. On the other hand, a major challenge is the reliance on expert opinions, which can be difficult to obtain and select. Moreover, these opinions can be conflicting and uncertain, as they are based on human judgment, which inherently involves subjectivity [3]. To overcome this issue, such methods also embed the calculation of a consistency index, which serves as a proxy for the logical coherence of judgments and helps ensure that the decision making process remains as reliable as possible [35].

Fuzzy Analytic Hierarchy Process (FAHP) Despite the use of a consistency index, the traditional AHP methods still faces limitations in handling the ambiguity and vagueness inherent in human judgment. Experts evaluations are often imprecise or linguistically expressed (e.g., “slightly more important” or “being far from the electrical grid is much less favorable than being far from a national park”), which makes it challenging to make precise judgments in pairwise comparisons. To better capture this uncertainty, the Fuzzy Analytic Hierarchy Process (FAHP) extends the conventional AHP by incorporating fuzzy set theory [40]. In a notable application of these principles, a study focusing on onshore wind farm site selection in South Korea [3] employed the FAHP method to determine factors weights more robustly. Similar approaches have also been used in other countries, such as Greece [29] and Nigeria [7].

Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) TOPSIS is a popular MCDM technique, originally introduced over four decades ago [50], and still remains (together with its variations) one of the most widely applied methods in multi-criteria analysis. The core idea is intuitive: the optimal alternative should be closest to the ideal solution, representing the best possible performance across all criteria, and farthest from the negative ideal solution, which represents the worst case scenario. By calculating the relative distances to these two reference points, TOPSIS provides a simple ranking of the alternatives. Its appeal lies in its simplicity and compatibility with a variety of decision making contexts. For example, in the context of wind farm site selection, the ideal location might correspond to high wind speed, a short distance to the grid, and low environmental impact. These criteria can be represented as a vector. Each site in the region of interest is also represented as a vector of values and then compared to the ideal and negative-ideal vectors. TOPSIS calculates the geometric distance of each site to both extremes and assigns a relative closeness score. The site with the highest score is considered the most suitable. TOPSIS allows for the inclusion of weights to reflect the relative importance of factors, adjusting the distance computations to the ideal and negative-ideal solutions. This is why techniques like AHP and TOPSIS are often used together. For example, in a study conducted in Turkey [14], a FAHP method was applied to determine the weights of the factors. These weights were then used in the TOPSIS framework to rank the alternatives. A similar idea was used in Eastern Macedonia and Thrace [28]. However, while TOPSIS offers an intuitive solution to the problem, it is not without limitations. A key concern arises during the data normalization phase, where vector representations are constructed. Since TOPSIS is highly sensitive to the way data is normalized, selecting an appropriate normalization method is critical. Poor choices at this stage can lead to distorted distances and, ultimately, unreliable rankings. This is particularly problematic when dealing with factors that have different units or highly skewed distributions. Recent studies have highlighted this issue [5, 10]. Additionally, TOPSIS during the distances caluclation assumes that criteria are statistically independent [34], overlooking potential interactions that may exist between them. Finally, the method relies on human input to define the ideal and negative ideal solutions, which introduces subjectivity into the process. As with Fuzzy AHP, fuzzy extensions of TOPSIS have also been developed to address this uncertainty [15].

The methods discussed so far, regardless of their specific type, commonly utilize Geographic Information Systems (GIS) to visualize the results. GIS refers to a system designed to capture, store, manipulate, analyze, manage, and present geographical data. In the context of wind farm site selection, GIS tools are essential for integrating various spatial datasets. These systems allow for the visualization of complex, multi dimensional data in a geographical context, making it easier to identify locations that are more suitable or less suitable. Each of the methods typically produces a ranking or a score for each location. This ranking or score can then be used as a form of legend on the GIS map. Several examples are provided in Figure 2.2.

2.2.2 Non-MCDM Methods

While MCDM methods offer structured frameworks for decision making, they rely heavily on subjective judgments to assign importance to the different factors. Making them inherently dependent on experts opinions and qualitative assessments. In contrast, Non-MCDM methods are predominantly fully data driven approaches. Although expert input

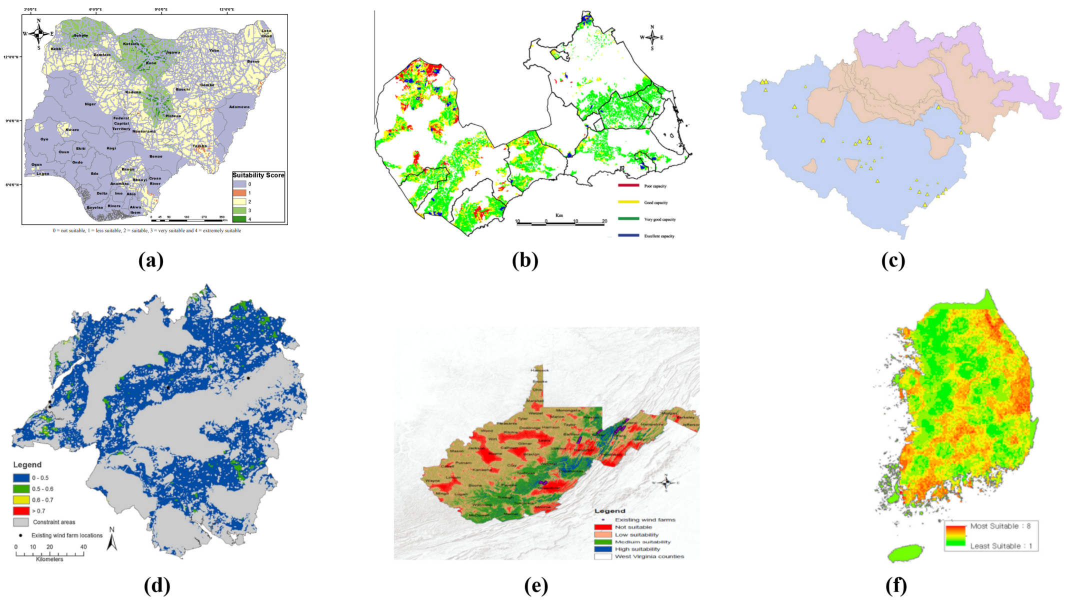


Figure 2.2: GIS-based visualizations showing the spatial distribution of site suitability for wind farm development. Regardless of the specific method used, each map represents suitability scores or rankings derived from multiple criteria.
(a) Nigeria, FAHP method [7] **(b)** Murcia region, Spain, FAHP method [42] **(c)** East Macedonia, Greece, AHP+TOPSIS method. Bigger triangles represent higher suitability [28] **(d)** South Central England, AHP method [33] **(e)** West Virginia, USA, AHP method [1] **(f)** South Korea, FAHP method [3]

may still be incorporated at certain stages, these methods fundamentally rely on computational models and algorithmic logic to derive conclusions. Within this broader category, AI models have emerged as a hot topic in recent years, due to their ability to handle complex, high-dimensional data and uncover patterns that traditional methods might miss. In the following section, we will discuss some examples of these Non-MCDM methods.

Cost/Benefit Models Unlike the MCDM methods described before, which emphasize a broader set of factors such as environmental, technical, and social criteria, cost/benefit models focus primarily on the economic aspects of wind farm site selection. These models are designed to evaluate the financial feasibility of different site options by comparing the costs associated with wind farm development and operation against the anticipated economic benefits, primarily in terms of energy production and long term revenues. In the study by Kim et al. [27] for example, the cost estimation for offshore wind farms involves a comprehensive breakdown of all economic factors. Each cost category is meticulously calculated, including:

- Turbine cost
- Foundation cost

- Electric system
- Grid interface cost
- Project development cost

The benefit estimation is also detailed, taking into account the annual energy production and efficiency loss. The potential locations are then ranked based on the cost/benefit ratio and visualized using GIS. Such an approach leaves very little room for subjectivity, as it relies heavily on quantitative data and objective calculations. Similarly, a study focusing on Turkey's wind energy potential [16] applies a similar cost/benefit approach to evaluate optimal sites. What is particularly interesting in this study is their use of the ARIMA (AutoRegressive Integrated Moving Average) model to forecast where the highest energy demand will be in the future. By predicting energy consumption trends, the researchers are able to strategically place turbines closer to areas where energy demand is expected to peak. This ensures that the wind farms are aligned with future energy needs, making the entire planning process more forward looking and efficient. Also in this case, the estimation is entirely data-driven, as a mixed-integer nonlinear programming model is used to minimize installation, maintenance, and operational costs while accounting for wind potential, demand forecasts, and budget limits across the Turkish provinces.

Supervised Machine Learning Algorithms Machine learning algorithms represent a powerful evolution of data-driven wind farm site selection methods, offering predictive capabilities and advanced pattern recognition that go beyond traditional statistical or optimization models. Several supervised learning models have been explored in this context. For instance, in a study conducted in the Masovian Voivodeship of Poland, an Artificial Neural Network (Multilayer Perceptron) was trained to evaluate land suitability for wind farm placement using spatial and topographic features [37]. The training data was created by dividing the region into $100\,\mathrm{m} \times 100\,\mathrm{m}$ land squares, each of which was evaluated by experts. Experts assessed each grid square and assigned a suitability score, which served as the target variable for training the model. The performance of the trained ANN showed strong predictive capability, with an accuracy value higher than 0.80 on the test set, indicating that the model was able to accurately capture the relationship between the input features and the land suitability. Similar good performance results were achieved using Support Vector Machines (SVM) and Random Forests (RF), as demonstrated by a study that employed global wind farm installation data [41]. Note that a common issue when using supervised learning methods is the imbalance in the dataset. In most cases, the labeled data, representing the actual locations where wind farms are placed or the optiaml ones, is significantly smaller than the set of all possible land locations. This class imbalance can cause the model to be biased toward predicting negative outcomes (unsuitable land). To address this issue, oversampling techniques like SMOTE (Synthetic Minority Over-sampling Technique) [13] are commonly used. SMOTE helps by generating synthetic samples for the underrepresented class, thus improving the model's ability to learn the characteristics of both classes more effectively and eventually obtain more accurate predictions. Other types of supervised models, such as Logistic Regression [22], are also used for similar purposes. However, we will not go into the specifics of these methods, as the underlying concept remains the same.

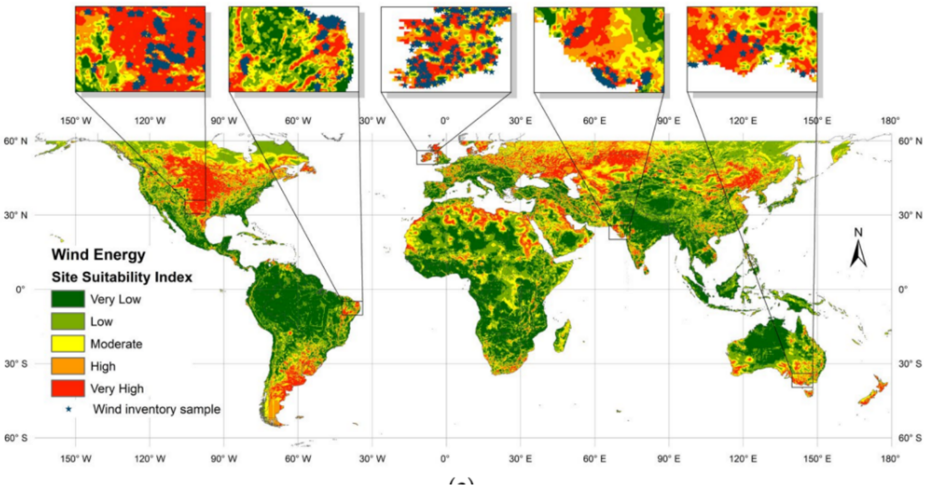


Figure 2.3: Global wind farm suitability map generated through supervised machine learning, trained on a dataset containing worldwide wind farm locations [41].

Unsupervised Machine Learning Algorithms Unsupervised Machine Learning offers an alternative approach to wind farm site selection that addresses some of the limitations associated with supervised learning models. Unsupervised techniques do not require ground truth data (labels). One critical issue with supervised learning in the context of our problem is the risk of model bias. When current wind farm locations are used as ground truth for training, the model basically learns to replicate the decisions made in the past, potentially reinforcing biases in site selection. This can result in the new model favoring areas similar to those where wind farms were previously established, even if those locations may not represent the most optimal set of choices. In other words, the model might automate a suboptimal decision-making process. If the supervised learning model is trained based on experts suggested locations (such as in [37]), instead of actual wind farm sites, it faces the same challenge previously discussed: human judgment. While experts can provide valuable input, their assessments can also be subjective.

The majority of unsupervised methods for wind farm location selection involve some form of clustering. A study demonstrated that Australian existing and planned wind farms produce less power and show greater energy output variability compared to a hypothetical optimal set of farms with the same capacity (even when the hypotetical set is constrained to locations within 100 km from the electrical grid) [21]. They arrived to this result by employing a clustering technique and by minimizing correlations in wind power potential between different sites. They applied hierarchical clustering techniques to analyze historical wind speed time series data. What they discovered was significant: the existing and planned wind farms are largely concentrated within the same clusters. This means that these farms are highly correlated (i.e. if one zone experiences low wind conditions, other nearby farms will likely face the same issue). This interdependence creates a problem and it is particularly crucial when considering the need to complement solar energy, especially since solar power is unavailable at night and relies on wind or other sources for consistent, reliable energy production. Hierarchical clustering is not the only algorithm employed in these studies. K-Means and DBSCAN are also popular choices, as demonstrated by another Australian study [38] and in Pakistan [4].

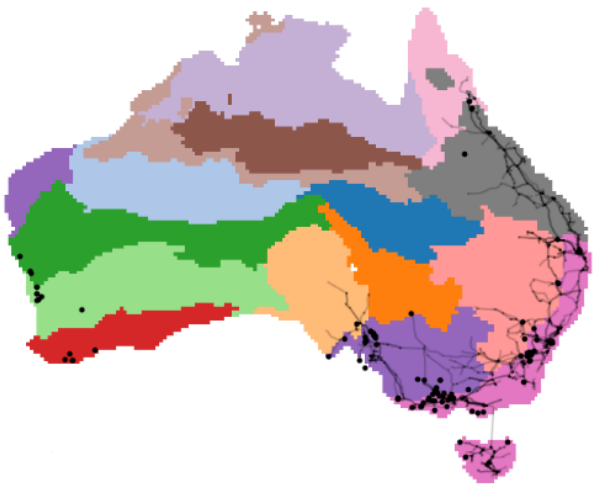


Figure 2.4: This figure illustrates the clustering of wind correlation zones based on historical wind speed data, showing that the majority of operating wind farms (dots) are concentrated within a few key clusters. This concentration is understandable, as these regions align with the locations of the electrical grid (lines) and Australia's population centers. However, the analysis also reveals that there are also better and optimal sites for wind farms within a 100 km radius from the grid, suggesting potential for reducing interdependence among farms, which could imporve energy reliability. [21].

2.3 Summary of the State of the Art

Current research on wind farm site selection includes a wide variety of methods and approaches, particularly within MCDM and AI-based techniques. Some of these were not even discussed in this document in order to comply with the report's length constraints.

However, in my opinion, a gap lies not in the availability of methods but in a critical lack of explainability and usability, especially from the perspective of non technical stakeholders such as citizens. This is not a secondary problem, because citizens are often directly impacted by the placement of new infrastructure, and tensions can arise if they do not fully understand the decision-making process or feel excluded from it. For instance, MCDM methods like AHP, FAHP, and TOPSIS are widely used and they heavily depend on expert derived factor weights and produce static GIS based maps. These outputs, while informative, do not support straightforward user interaction or allow non experts to explore alternative "what-if" scenarios based on different preferences or priorities. Few, if any, studies have incorporated dynamic interfaces that would allow users to adjust factor weights and immediately visualize how those changes affect site suitability outcomes. Introducing such tools could greatly increase both inclusivity and public trust. This is even more true for methods that fully rely on AI models and do not involve expert derived decisions. While explainability has been (and is) extensively studied in the field of Explainable AI (XAI), its application within the context of wind farm site selection is still relatively immature. Only a few recent papers, such as [12,47], have begun to address this gap, and even then, the integration of explainability is still not presented in an interactive interface format. This is unfortunate, as GIS systems and XAI techniques are well suited for integration, as demonstrated in recent studies such as [31]. Moreover, while wind farm site selection has been extensively studied globally, Australia remains underrepresented in the literature. Wimhurst et al. in their comprehensive review of wind farm site suitability models up until 2022 [55], noted that no studies had specifically addressed this topic in the Australian context (Figure 2.5). Since then, a few papers have emerged, such as those referenced earlier (e.g., Rahimi et al. [38], Gunn et al. [21]), yet the volume of research remains limited. Especially given Australia's substantial wind resources, there is considerable potential for further exploration in this area.

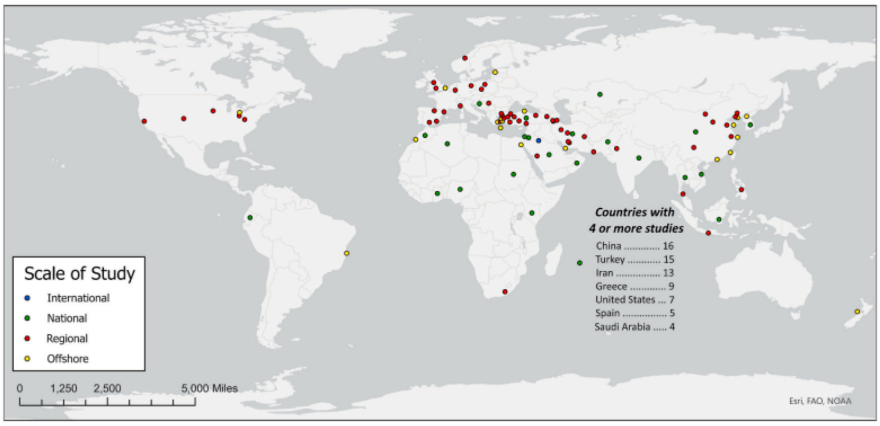


Figure 2.5: Study locations of articles on wind farm site suitability models published up to 2022 [55]. Note that there were no studies focused on Australia.

2.4 Research Project Justification

As discussed in the Summary of the State of the Art, in my opinion, there is a clear gap in the current body of research regarding the explainability and interactivity of wind farm site selection methods, especially from the perspective of non technical stakeholders. Although many advanced planning techniques exist, most of them are understandable only to experts with deep domain knowledge, making it hard for the general public to engage with or trust the outcomes. Thus, this research aims to help close that gap by developing a simple, interactive tool that makes wind farm site selection more transparent and easier to understand. The goal is to build a GIS based interface where users, whether they're citizens or planners, can explore possible locations, adjust the importance of different factors, and clearly see how those changes affect the results. Instead of creating a brand new method, the focus is on taking existing techniques and presenting them in a way that's more open and user friendly. In summary, the ultimate goal is to improve communication and understanding of how decisions are taken in wind farm planning, because often they are not very well explained. To illustrate this point, the figures 2.7 and 2.6 below show two examples of wind farm developments in Victoria where the reasoning behind site selection could have been better communicated. And these aren't isolated cases. There are plenty more where the decision-making process could have been explained much more clearly. Such a tool could be used, for instance, alongside more lengthy government or industry documents like the Integrated System Plan For the National Electricity Market (2024) by AEMO [6]. To explain long-term planning, helping to break down complex information and make it more accessible.

For Public Notice via Internet

REASONS FOR DECISION UNDER *ENVIRONMENT EFFECTS ACT 1978*

Title of Proposal: **Newfield Wind Farm**

Proponent: **Acciona Energy Oceania**

Description of Project:

Up to 15 turbines of about 1.5 MW installed capacity would be installed along a 4km ridgeline at Newfield, approximately 7.5 km north of Port Campbell. The total installed capacity of the wind farm would be about 22.5MW. The wind farm would include wind turbine generators, underground cabling, an electrical substation (transformers, switchyard and control building), access tracks, crane pads and anemometer.

The attached map and aerial photograph shows the location of the works.

Decision:

The Minister for Planning has determined that an **Environment Effects Statement is not required** for this proposed wind farm.

Reasons for Decision:

- The project site mostly consists of cleared agricultural land, with considerable scope to adjust the siting of turbines and associated infrastructure for this wind energy facility to avoid significant adverse effects on indigenous flora and fauna and to mitigate other environmental effects.
- Potential effects on landscape values and residential amenity are likely to be of no more than local significance.
- The planning permit process, informed by the *Policy and planning guidelines for development of wind energy facilities in Victoria,* offers a suitable mechanism to assess potential environmental effects and related issues.

Figure 2.6: Reasons for the site selection of Newfield wind farm (Victoria), available to the public. Retrieved from: Newfield Wind Farm – Planning Referral.

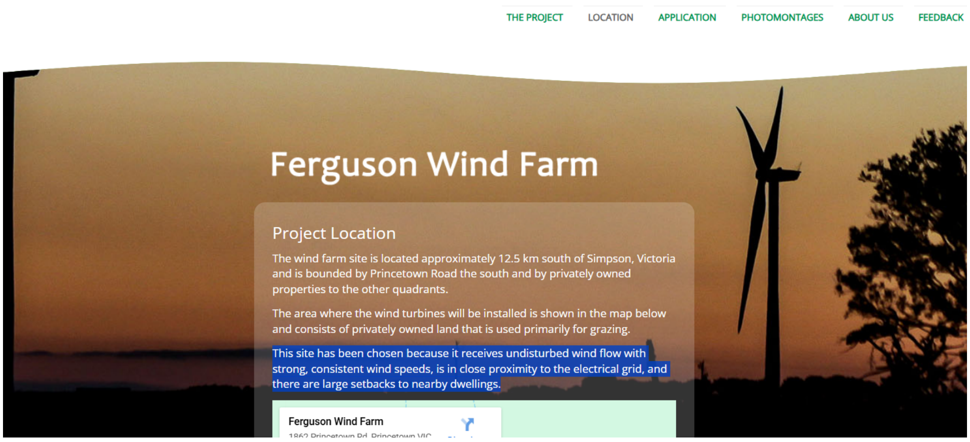


Figure 2.7: Ferguson Wind Farm – Project Location. Another example of poorly communicated reasoning behind the selection of a wind farm site.

Methodology

The following project plan outlines the general approach used to build the proposed site selection tool. While the steps below are laid out as distinct phases for the sake of explanation and presentation, the process was flexible. As the project developed, certain tasks shifted, overlapped, and changed based on what was learned in the way. The aim was to move forward in manageable stages, with space to experiment, gather feedback, and refine the tool.

1. **Define the criteria and collect data**
   The first step was to decide which factors to include in the site selection process, guided by existing research and case studies. Once the criteria were finalized, the necessary spatial data, specific for Australia, were gathered from open access sources. Probably the biggest challenge at this point was the harmonization of datasets from different providers, in terms of format, resolution, and coverage. This stage was very important to ensure a consistent foundation for the system. Since the primary focus of this project was on building a visual interface, the number of criteria used at that stage was fewer than in more comprehensive studies. However, the structure remained open and extensible, allowing additional factors to be incorporated in the future. That said, several core criteria, such as wind speed, distance to the electricity grid, proximity to national parks, were definitely included from the very beginning. Additionally, an extra consideration taken to increase feasibility was narrowing the scope to focus exclusively on onshore turbines. Since the data used for this project is open source (e.g. ERA5 global reanalysis for weather data [23], electricity infrastructure locations) and does not involve human subjects or personal information, ethical concerns related to privacy or data protection were not needed.

2. **Set up the basic GIS framework**
   Instead of using classical GIS tools, I used web based mapping frameworks that are open-source and easy to deploy. This approach allows for greater flexibility and accessibility. Additionally, I used Python for the backend of the interface,

allowing the system to more smoothly integrated with certain libraries. A suitable framework for building this setup was found to be Plotly Dash, which provides both the interactivity needed for the frontend and the ease of use for the backend integration. The focus at this stage was on loading and visualizing the collected data for analysis. No logic was implemented yet.

3. **Add the logic**
   A simple decision-making method, such as a weighted overlay to generate a suitability map.

4. **Make it interactive**
   An interactive interface was developed, allowing users to adjust weights and settings in real-time and instantly observe how the results change. This is the crucial functionality for making the tool both accessible and engaging. It is important to note that the optimal weights, typically required for techniques like TOPSIS or AHP, were not calculated as part of this tool, as this would have required expert judgment. Instead, the goal was to enable users to visualize how the suitability map changes based on any set of preferences or weights they choose, regardless of their optimality or accuracy.

5. **Bring in explainabilty**
   This section can be viewed from three stages of explainability, each contributing to improving transparency.

   - **Improving the weight-based system**
     The weights based system previously implemented was inherently self explanatory, as it allowed users to adjust the importance of various factors and see the direct impact on the suitability map. To further enhance explainability, additional features were integrated. For instance, if a user clicks on a specific location on the map, the interface shows how far that location is from the electricity grid (e.g., 300 km). It also displays how the 300 kms compares to the distribution of distances for all points, providing users with context on whether this distance is relatively short, long, or within an expected range. The distance from the grid is used as an example here, but the same can be applied for all the different factors. For example, a very good illustration (and insipration) of this principle can be found in the Australian Cancer Atlas. In their interactive mapping tool, users can explore various health data layers and click on specific locations to view contextual information, such as how a particular area compares to the rest of the nation.

- **Integrating an AI model for site suitability**
  In this stage, a machine learning model was implemented to generate an additional suitability map. While the model was far from being perfect, especially given the challenges posed by a very imbalanced dataset (with only 100-200 actual or planned wind farm locations compared to 12,000 possible onshore locations across Australia, when for example gridded at 0.25° latitude-longitude resolution), the focus here was to explore the potential for integrating XAI techniques into the tool. The suitability map produced by the model could be improved with explanations regarding the different features importances. The aim is to demonstrate that, independently of achieving high model performance, there is value in visually presenting explainability.
- **Exploring the sub-optimality of the current Australian wind farms configuration**
  This part of the project researched on how clustering techniques can be made more interactive. The goal was to help users understanding the spatial patterns and optimization potential in wind farm placement. For example, the previously cited paper [21] suggests that the current configuration of wind farms in Australia is suboptimal, yet the results presented are static and rely on a fixed, hard coded number of clusters or wind zones. This was improved.

6. **Test with users. refine and adjust**
   The tool was shared with a group of users to observe how they interacted with it. What was intuitive, what was confusing, and where things could be better. Based on feedback, both the interface and the underlying system were improved.

Each of these steps plays a critical role in the development of the tool. Throughout the following parts, specific features and implementation choices will be referenced according to the corresponding numbered stage above.

3.1 Variables Used in the Site Selection Tool

This section addresses the first step outlined in the methodology, namely the definition of criteria and the collection of data. The site selection process for onshore wind farms typically involves a large number of environmental, technical, and regulatory variables. However, as explained before, the primary aim of this project was not to create the most exhaustive or data-rich tool, but rather to demonstrate how variables can be combined in a way that is transparent, interactive, and understandable. To support this, a deliberately limited set of variables was selected. The tool was designed to show how users can explore trade offs and decisions by adjusting factor weights and receiving real time, contextual feedback. Additionally, the project introduces a novel variable that, to the best of current knowledge, has not been used in prior siting studies: a metric designed to capture the correlation of potential sites with existing wind farms. Below is a brief explanation of the variables implemented in the tool:

Distance to Electrical Grid The distance from a potential site to the existing electrical grid is one of the most critical technical variables in wind farm siting. Connecting a wind farm to the transmission network involves substantial infrastructure costs, regulatory approvals, and construction time. As a result, sites that are located far from the grid are often less economically viable. In this project, the distance to the grid was calculated as the straight line distance (in KMs) from each point in the analysis grid to the nearest transmission line. The necessary spatial data were obtained from the open source Digital Atlas provided by the Australian Government.

Wind Capacity Factor For each location in Australia, this variable represents the average wind capacity factor, estimated over a two years period. The capacity factor is a dimensionless value (ranging from 0 to 1) that indicates the proportion of a wind turbine's rated power output that is actually produced over time. It is a more meaningful and practical metric than raw wind speed, as it directly reflects the expected energy yield from a given site.

Wind speed must be translated into energy output using a turbine specific power curve, which defines how a turbine converts varying wind speeds into electricity. The typical power curve begins at a cut in speed, below which the turbine does not generate power. Between the cut-in and rated speeds, power output increases roughly with the cube of wind speed. Above the rated speed, the output plateaus until a cut out speed is reached, for safety reasons, at which point the turbine stops operating. To calculate capacity factor from wind speed, this project applies a smoothed composite function based on standard turbine performance characteristics, see figure 3.1.

The data source for the wind speed measured at 100 meters above ground level is the ECMWF Reanalysis 5th Generation (ERA5) dataset

Distance to Nature Land The distance to protected natural areas was calculated similarly to the distance to the electrical grid. However, unlike grid proximity, which is more favorable when minimized, this variable is ideally maximized, as greater distance from sensitive environmental zones reduces potential ecological impact. The data source for this variable is the open source Australian Protected Areas Database.

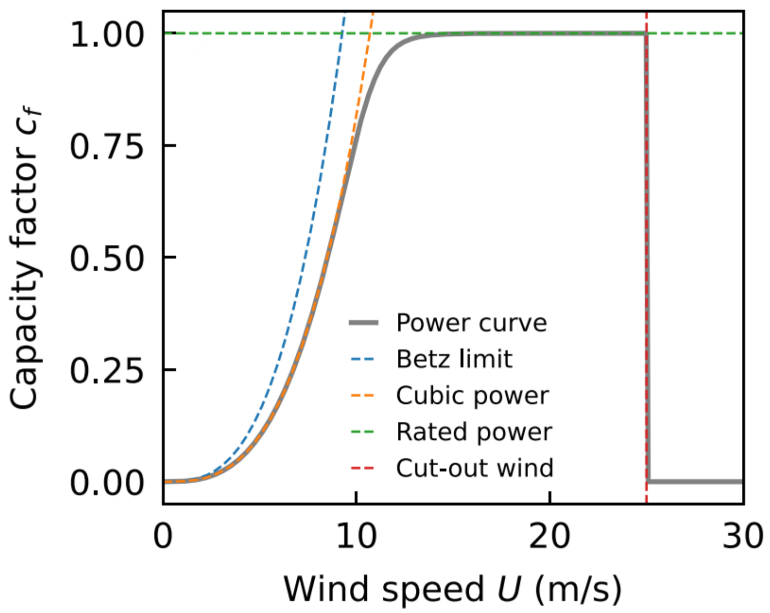


Figure 3.1: Standard power curve (grey) used to obtain wind capacity factors [21].

Correlation with Existing Farms This variable quantifies the degree to which the wind profile at a candidate location correlates with those of existing wind farms across Australia. For each location, we calculate the average correlation between its wind speed time series and those recorded at operational wind farm locations. A high correlation indicates that wind patterns at the new site closely mirror those of existing farms, causing their energy outputs to fluctuate simultaneously. Such synchronized generation can increase variability and reduce the overall stability of the electricity supply. Therefore, minimizing this correlation is desirable, as it encourages spatial diversification of wind resources, leading to a more balanced and reliable energy production across the grid.This is a novel variable introduced in this study, inspired by [21], and, unlike the previous criteria, it has not been applied in previous wind farm siting research. By incorporating this measure, the tool aims to address inefficiencies associated with clustering wind farms in highly correlated wind regimes. To compute this variable, a wind correlation (symmetric) matrix was constructed, where each entry represents the Pearson correlation coefficient between the wind speed time series of two locations.

Below is an example of this process. In this simplified toy example, we consider two potential new wind farm locations: one in Sydney and one in Melbourne. The existing wind farms, represented as *WindFarm1* and *WindFarm2*, are already operational sites.

While in the real implementation the locations are defined at a much finer spatial resolution (e.g., gridded points across the country), this illustrative matrix helps demonstrate the logic behind the correlation-based suitability assessment.

$$
\begin{array}{c c}
 & \begin{array}{cccc} Melbourne & Sydney & WindFarm1 & WindFarm2 \end{array} \\
\begin{array}{c} Melbourne \\ Sydney \\ WindFarm1 \\ WindFarm2 \end{array} &
\begin{pmatrix} 1 & 0.21 & \color{red}{0.42} & \color{red}{0.86} \\ 0.21 & 1 & \color{blue}{0.26} & \color{blue}{0.32} \\ 0.42 & 0.26 & 1 & 0.13 \\ 0.86 & 0.32 & 0.13 & 1 \end{pmatrix}
\end{array}
$$

For each candidate site, Sydney or Melbourne, the average correlation with existing wind farms is computed (average of red and blue entries in this case). The average could also be a weighted average, where the weights correspond to the total installed capacities of the existing wind farms.

$$
\begin{array}{c c}
 & AverageCorrelation \\
\begin{array}{c} Melbourne \\ Sydney \end{array} &
\begin{pmatrix} \color{red}{0.64} \\ \color{blue}{0.29} \end{pmatrix}
\end{array}
$$

A lower average correlation (Sydney in this example) indicates a more diversified wind profile, which is preferred in order to enhance the overall energy production stability and reduce synchronized variability. In other words, if one day it is not windy in an area where many farms are located, the overall energy production will be low due to the high correlation in their wind patterns.

Solar Radiation Solar radiation is measured as surface irradiance (W/m^2) and is included to demonstrate how the same environmental variable can be interpreted either as a benefit or a cost when choosing the ideal location, depending on the specific development goals.

- **Cost:** High solar irradiance may indicate that a site is more suitable for photovoltaic (solar) energy development rather than wind energy. In this context, the variable is something we want to minimize. Thus, locations with high solar radiation will have lower suitability scores.

- **Benefit:** Conversely, if the objective is to promote co-location of wind and solar farms, high solar irradiance becomes an advantage, increasing the overall suitability of the site.

This flexible treatment allows the tool to adapt to different planning priorities. By default, the variable is considered a cost. The data source used for computing mean solar irradiance is ECMWF Reanalysis 5th Generation (ERA5) dataset.

3.2 Locations

As part of step 1 of the methodology, which involved defining the site selection criteria and collecting spatial data to evaluate potential wind farm locations across Australia, the study area was discretized using a regular grid with a resolution of 0.25° in both latitude and longitude. This resolution was chosen to match that of the ERA5 dataset, which was used as the primary source of wind speed data. Aligning the spatial resolution of the analysis with that of the wind data ensures consistency and avoids downscaling/upscaling errors. Each grid point was treated as a potential site and evaluated across all selected criteria (e.g., wind capacity factor, distance to grid). This discretization resulted in 11,822 location points being analyzed across the Australian onshore area. Since no interpolation was applied to smooth the data between grid points, each location is effectively represented as a discrete point on the map. As a result, when the suitability map is viewed at a national scale, the output appears smooth and continuous due to the density of the grid. However, when zooming into specific regions, the underlying grid structure becomes visible, as shown in Figure 3.2. This visual effect is a direct consequence of the chosen resolution and the decision not to apply spatial interpolation.

Interpolation was intentionally omitted in this project, as the primary objective was to develop a tool that effectively conveys the underlying conceptual approach from a qualitative perspective. Should the tool be applied to a more localized area, or intended for operational use, interpolation techniques or data sources with finer spatial granularity would be necessary. Further discussion on this limitation and its implications can be found in the Future Work section.

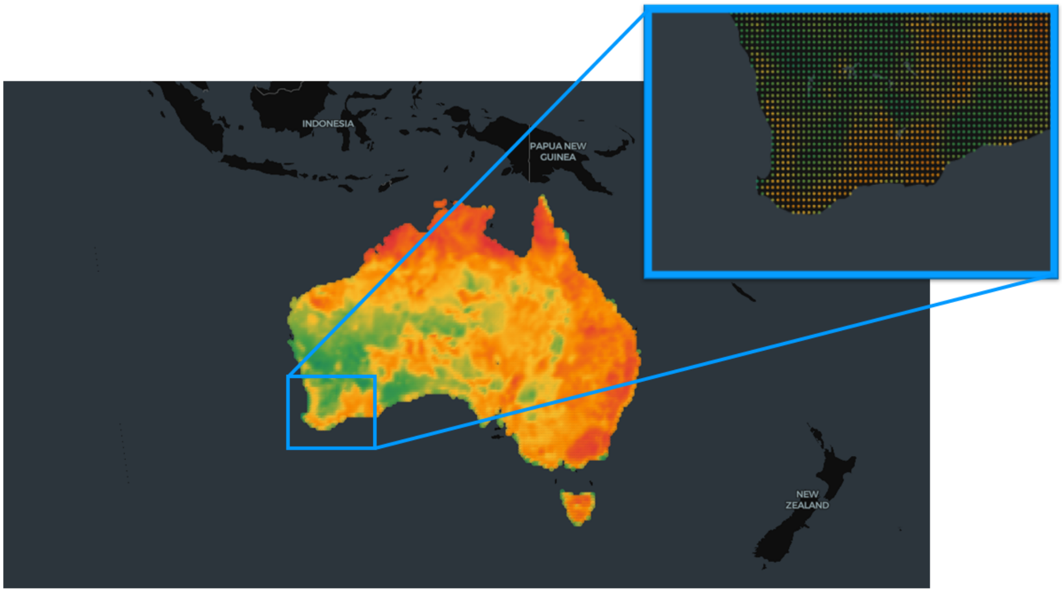


Figure 3.2: Zoomed-in view of the suitability map showing the underlying 0.25° longitude/latitude grid structure. While the map appears smooth at the national scale due to the density of data points, the discretization becomes visible at finer scales. This is a direct result of the chosen resolution and the absence of interpolation

3.3 Suitability Index

This section aligns with steps 2, 3, and 4 of the methodology, setting up the GIS framework, adding decision logic, and enabling interactivity. The Suitability Index is the core feature of the platform and provides a single, interpretable metric that quantifies how appropriate a given location is for wind farm development. It is designed to combine multiple decision factors into a unified score, while maintaining full transparency and user control.

The index is calculated as a weighted linear combination of the normalized input variables described in the previous sections. Users assign custom weights to each variable through an interactive interface (e.g., sliders or input fields), allowing them to prioritize the factors that matter most for their specific planning objectives. The sum of all weights is constrained to equal 1, ensuring a balanced and interpretable aggregation. Given that the input variables vary significantly in terms of units, ranges, and statistical distributions, a normalization step is applied to enable fair comparison and enhance explainability. Each variable is transformed into a standardized score on a 0–100 scale using percentile ranking, where 100 always represents the most favorable value (i.e., highest suitability) and 0 the least favorable.

Using percentiles was chosen as the most appropriate approach for this project because it allows for the creation of a single informative score for each variable without introducing any prior assumptions about their relative importance or distribution. This ensures that the scaling remains consistent and fully data driven. The figure 3.3 (taken from the tool's documentation) visually illustrates this normalization process, showing how raw values are converted into standardized scores based on percentiles and linearly combined to create the suitability index.

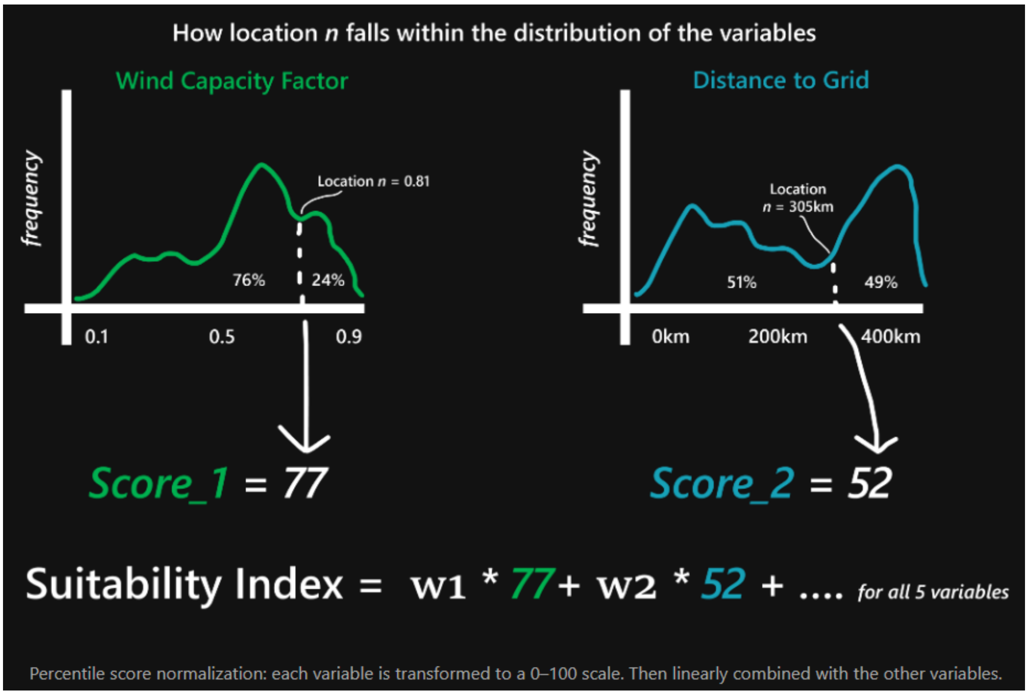


Figure 3.3: Percentile based normalization (image taken from the the tool's documentation). In this example, if a location is closer to the electrical grid than 76% of other locations, it will receive a score of 77/100 for the 'Distance to Grid' metric. This transformation allows for consistent comparison and clearer interpretations. The final suitability score is then computed as a weighted linear combination of these normalized values, where the weights are defined by the user through the interactive interface.

To further illustrate the Suitability Index principle, two screenshots from the developed tool are provided below. In the first example (Fig. 3.4), non-zero weights were assigned to all variables. The resulting map displays overall suitability, where greener areas indicate higher suitability for wind farm development. The second example (Fig. 3.5) represents an edge case, where all the weight is assigned solely to the 'Distance to Electrical Grid' variable. As expected, the resulting map essentially reflects a direct overlay of the grid infrastructure, highlighting areas closest to the network as the most suitable.

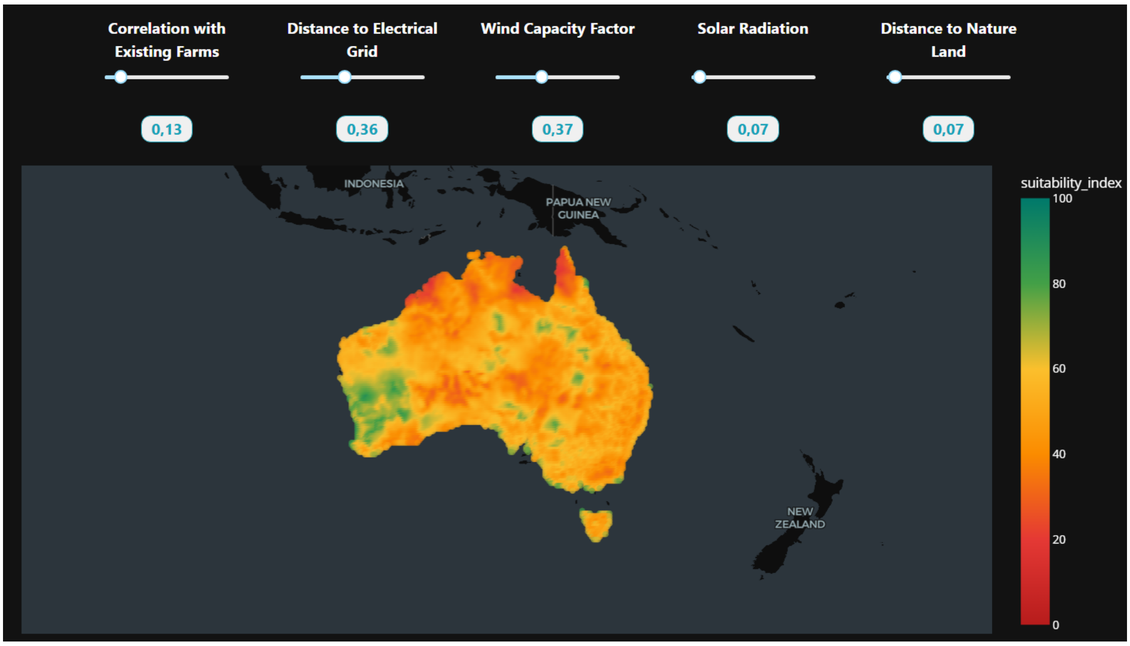


Figure 3.4: Suitability map generated by assigning non-zero weights to all variables, resulting in a comprehensive suitability index. Greener areas indicate higher suitability for wind farm development.

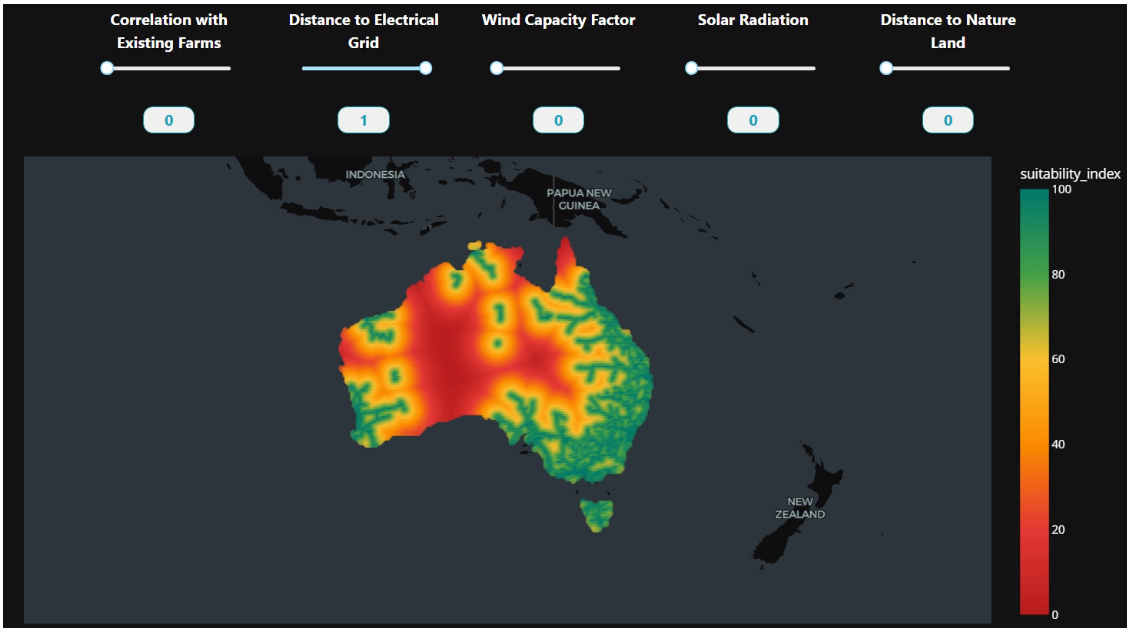


Figure 3.5: Suitability map generated by assigning all the weight to the 'Distance to Electrical Grid' variable (slider set to 1, all others to 0). Note that the map essentially represents an overlay of the grid infrastructure, highlighting areas closest to the network as the most suitable.

3.4 Backtesting of the Suitability Index

To evaluate the effectiveness of the proposed Suitability Index, a backtesting procedure was conducted using historical data and known wind farm developments. The weights assigned to each variable in the Suitability Index were set as follows:

- Distance to Electrical Grid: 0.40
- Wind Correlation with Existing Farms: 0.30
- Wind Capacity Factor: 0.20
- Solar Radiation: 0.05
- Distance to Protected Nature Land: 0.05

The backtesting process involved analyzing all new wind farms constructed in Australia after 2015. For each of these farms, the relevant variables for the Suitability Index, such as average capacity factor and correlation with existing farms, were computed using data from the ten years prior to their construction (e.g., for a wind farm built in 2016, data from 2005 to 2015 were used).

Using these historical variables and the specified weights, the location with the highest Suitability Index score was identified for each year before construction. This "most suitable" location was then compared against the actual site where the wind farm was built in subsequent years.

The comparison focused on the surplus in hourly capacity factor between the top-ranked location according to the Suitability Index and the real development site. Results consistently showed that the locations identified by the Suitability Index would have yielded higher energy production, demonstrating that a top-down site selection approach, that explicitly accounts for factors like wind correlation, can be more efficient than the traditional bottom-up approach.

To ensure a fair comparison, an additional constraint was applied whereby the selected "most suitable" locations had to be within 100 km of the electrical grid. Even under this constraint, the Suitability Index locations consistently outperformed the actual sites, confirming the robustness of the method. It is important to note that these results were obtained using weights assigned somewhat intuitively but plausibly, without performing any formal optimization to identify the best possible weight combination. This indicates that even a simple top-down approach to site selection, rather than a purely bottom-up one, can lead to more efficient wind farm placement and improved energy production outcomes.

3.5 Pareto Efficiency

This component of the tool aligns with step 4 of the methodology, which focused on enhancing interactivity. By introducing Pareto efficiency analysis the platform offers a more dynamic and exploratory experience. In multi-criteria decision-making problems such as wind farm site selection, trade-offs between competing variables are common. For example, locations with high average wind capacity factors may be far from the electrical grid, increasing infrastructure costs, while sites close to the grid might have lower wind potential (3.6). In such cases, no single solution optimizes all criteria simultaneously. Instead, a set of optimal compromises exists, known as the Pareto frontier. Pareto optimization identifies these non-dominated solutions, points where improving one criterion would degrade another (Algorithm 1). The algorithm can be easily extended to identify multiple layers or "tiers" of Pareto optimal points beyond the first frontier. These Pareto tiers represent successive groups of non-dominated solutions after removing the previous frontiers. For example, the first tier contains the most optimal locations that cannot be improved in any criterion without sacrificing another, while the second tier includes points that are optimal once the first tier is excluded, and so on.

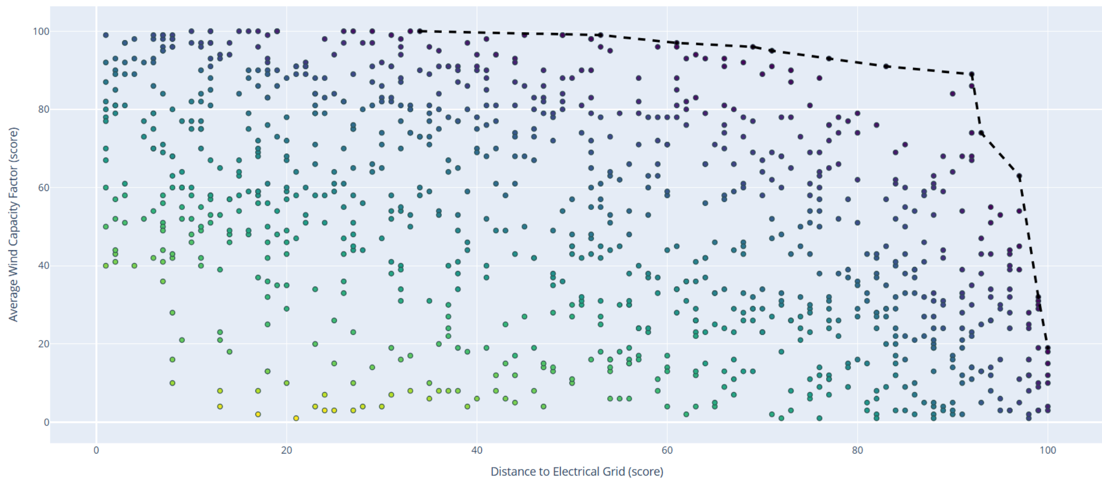


Figure 3.6: Visualization of the Pareto frontier in 2D for a subset of the location points. It illustrates the trade-off between wind capacity factor and distance to electrical grid. Each point represents a location scored on both criteria, with higher scores indicating better suitability. Points near the origin correspond to low scores on both variables (less suitable), while points on the Pareto frontier (dotted line) represent optimal trade-offs that cannot be improved in one criterion without worsening the other.

A key advantage of applying Pareto optimization to geographic data is that each Pareto optimal point corresponds to a physical location, allowing the results to be directly mapped onto a 2D spatial plane defined by latitude and longitude. This facilitates intuitive visualization and spatial analysis of trade-offs across multiple criteria.

Algorithm 1 Identification of Pareto Optimal Points

Require: Set of points $P = \{p_1, p_2, ..., p_n\}$, where each $p_i = (s_{i1}, s_{i2}, ..., s_{im})$ is a vector of normalized scores for m variables (higher values are better)

Ensure: Subset $P^* \subseteq P$ of Pareto optimal points

Initialize $P^* \leftarrow \emptyset$
for each point $p_i \in P$ **do**
  $dominated \leftarrow$ False
  for each point $p_j \in P$, with $j \neq i$ **do**
    if ***Dominates***(p_j, p_i) **then**
      $dominated \leftarrow$ True
      break
    end if
  end for
  if not $dominated$ **then**
    Add p_i to P^*
  end if
end for
return P^*

Dominates p_a, p_b {Returns True if p_a dominates p_b}
$better\_or\_equal \leftarrow$ True
$strictly\_better \leftarrow$ False
for $k = 1$ to m **do**
  if $s_{a,k} < s_{b,k}$ **then**
    $better\_or\_equal \leftarrow$ False
    return False
  else if $s_{a,k} > s_{b,k}$ **then**
    $strictly\_better \leftarrow$ True
  end if
end for
return $better\_or\_equal \wedge strictly\_better$

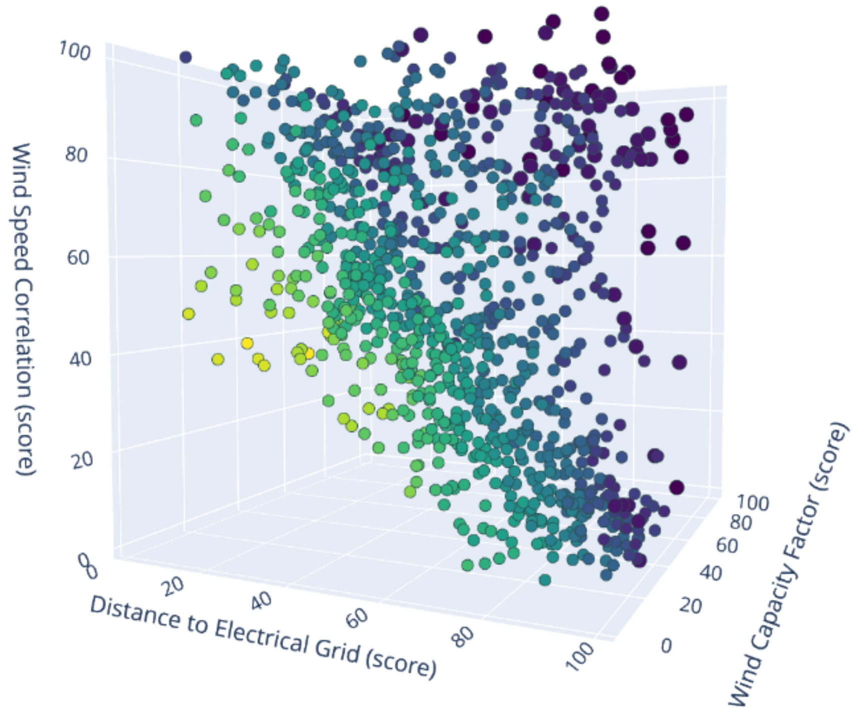


Figure 3.7: 3D visualization of the Pareto frontier (purple points) based on three variables: average wind capacity factor, distance to electrical grid, and wind correlation with existing farms. The Pareto optimization algorithm can process an arbitrary number of variables, identifying non-dominated solutions in multidimensional space. Since each point corresponds to a geographic location, these optimal solutions can be projected back onto the 2D map using latitude and longitude, facilitating clear spatial interpretation of trade-offs.

The Pareto optimization feature in the tool includes a slider that allows users to select which Pareto tier to display on the map, where tier 0 corresponds to the Pareto-optimal points (the first frontier) and tier 14 includes all points (the least optimal). After filtering by Pareto tier, users can further adjust the suitability index weights via sliders to dynamically explore how the overall suitability changes (3.8). This interactive process effectively allows users to navigate along the Pareto frontier in geographic space, visualizing in real-time which locations best balance different trade-offs between variables. While the algorithm treats all points on the frontier as equally optimal, since none can improve in one criterion without worsening another, not all trade-offs are equally desirable in real world planning. For example, being closer to the electrical grid may be more valuable than achieving a marginally higher wind capacity factor. This is precisely where the interactive weighting system becomes powerful: it lets users inject domain-specific knowledge or preferences into the decision process, helping them identify the Pareto-optimal sites that best align with practical development goals.

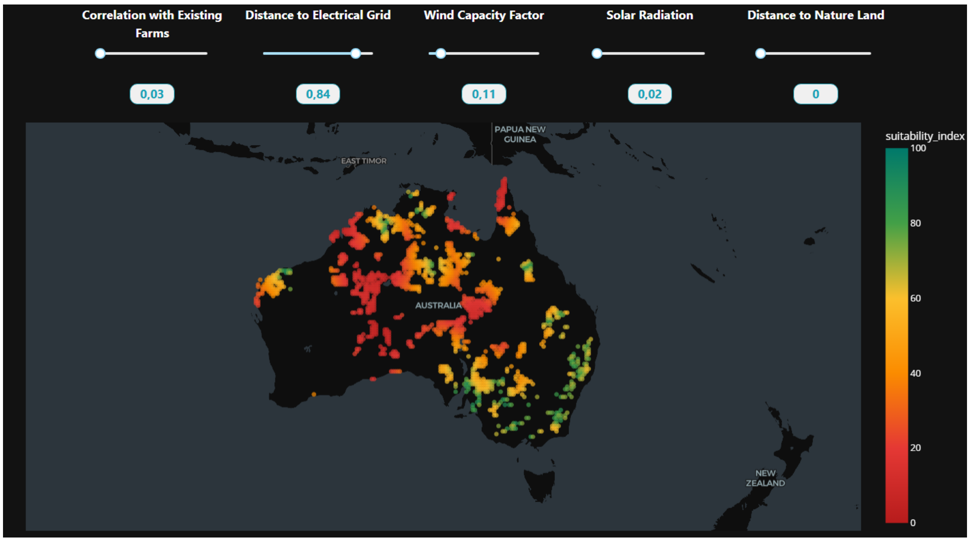


Figure 3.8: Changing the weights of the suitability index after filtering the Pareto optimal points enables users to move dynamically along the Pareto frontier. This interaction helps identify which Pareto optimal points, considered equally optimal on the frontier, best optimize specific trade-offs among the variables.

3.6 Interface Features

This section aligns with step 5 of the methodology, which focuses on explainability. To make the decision making process more transparent, the tool incorporates interactive features that help users not only identify suitable locations, but also understand the reasoning behind their suitability. These features are designed to support both exploration and interpretation by providing clear, contextual insights into how each location performs across the different evaluation criteria.

One key feature allows users to click on any point on the map, opening a detailed view of that specific location. This secondary panel provides both a statistical and visual breakdown of how the selected point performs across all variables included in the suitability index. It is possible to see the individual suitability scores (from 0 to 100) for each of the five variables used in the index (Fig. 3.9c), the absolute values of these variables and a contextual distribution plot for each variable, showing where the selected point falls within the full distribution across all locations (Fig. 3.9b). This allows users to interpret, for example, whether a site's capacity factor is relatively high or low compared to the rest of the dataset. Users can filter locations according to Paretro tiers (as explained in the previous section) and based on geographical constraints, such as limiting the display to specific states as well as filtering by setting suitability index thresholds (Fig. 3.9d). All of the dynamic filters can be combined together.

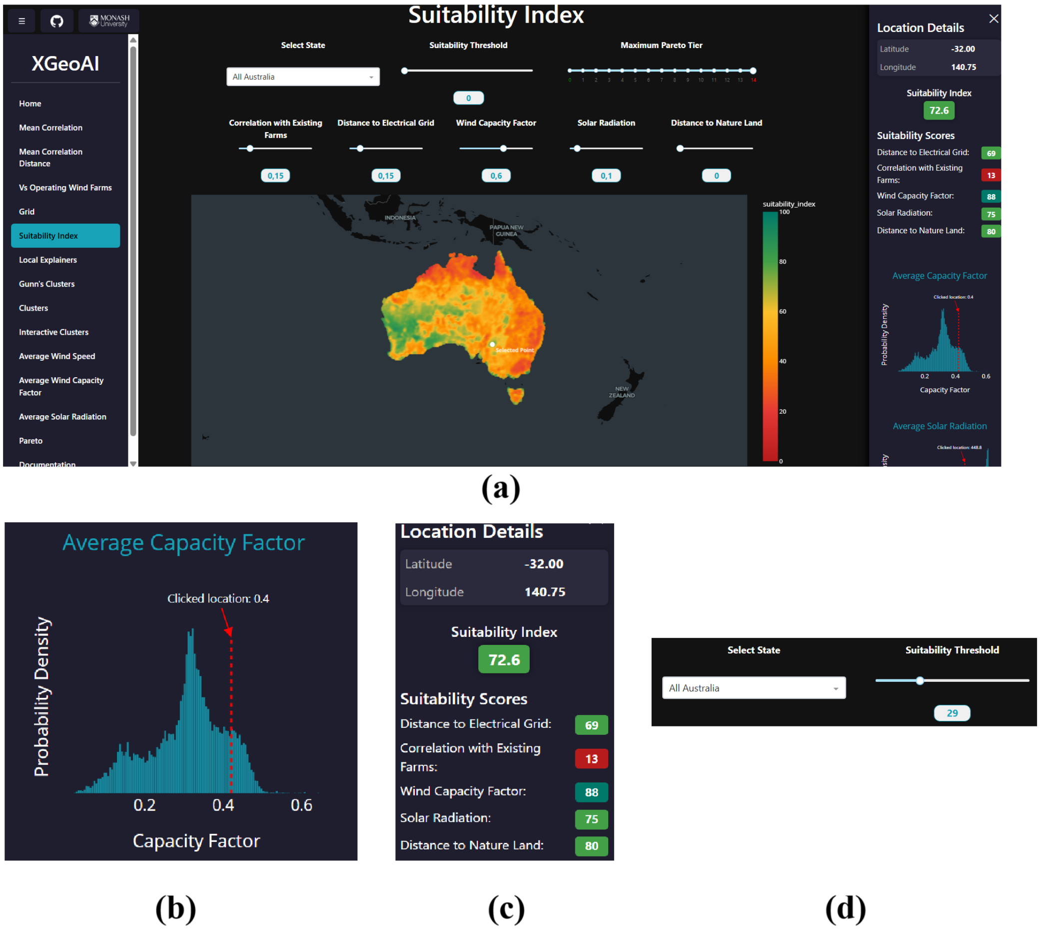


Figure 3.9: Overview of some interactive features in the tool. (**a**) Map view of all potential wind farm locations with color indicating suitability index according to the chosen weights (**b**) Zoom-in on the contextual distribution plot showing where a selected point falls within the dataset (there is one plot for each variable) (**c**) Detailed panel displaying suitability scores (0–100) for the five variables used in the index. (**d**) Interface controls allowing dynamic filtering by Pareto tier, geographic region (state), and suitability index thresholds. These interactive features help users explore trade-offs and assess individual sites in depth.

3.7 Clustering

This section aligns with step 5 of the methodology, focusing on explainability. In particular, it aims to visually highlight the sub-optimality of the current configuration of operational wind farms in Australia, as discussed in recent research [21]. While much of the tool is centered on evaluating new potential sites, this component shifts the focus toward understanding how existing sites are distributed spatially, and how their current layout may not fully exploit the available wind resources.

To support this analysis, clustering techniques were applied to group regions exhibiting similar wind behavior based on a constructed correlation matrix. The aim was to identify zones with comparable wind characteristics and then assess how existing turbines are distributed across these zones. The analysis reveals that, independently of the clustering technique used, operational wind farms tend to be concentrated within only a few clusters, indicating a lack of spatial sparsity in their placement. While this pattern is expected due to Australia's population distribution, being densely populated in some regions and very sparse across much of the territory, applying wind clustering techniques provides quantitative measures that illustrate this concentration and the resulting spatial inefficiency.

In the following section, we compare the outputs of different clustering methods. To perform a fair analysis all these algorithms are based on the same underlying data source: the wind correlation matrix. This correlation matrix is constructed by calculating the Pearson correlation coefficients between the wind speed time series (over 10 years) of locations across Australia. Each entry represents the correlation in wind behavior between two locations. High correlation indicates similar wind patterns, which can lead to synchronized energy fluctuations.

$$\begin{array}{c|c} & \begin{array}{ccccc} \text{Loc 1} & \text{Loc 2} & \text{Loc 3} & \text{Loc 4} & \text{Loc 5} \end{array} \\ \begin{array}{c} \text{Loc 1} \\ \text{Loc 2} \\ \text{Loc 3} \\ \text{Loc 4} \\ \text{Loc 5} \end{array} & \begin{pmatrix} 1.00 & 0.67 & 0.20 & 0.40 & 0.35 \\ 0.67 & 1.00 & 0.39 & 0.55 & 0.53 \\ 0.20 & 0.39 & 1.00 & 0.15 & 0.25 \\ 0.40 & 0.55 & 0.15 & 1.00 & 0.42 \\ 0.35 & 0.53 & 0.25 & 0.42 & 1.00 \end{pmatrix} \end{array}$$

Figure 3.10: Example of a wind speed correlation matrix, where each entry represents the Pearson correlation coefficient between the wind speed time series of two different locations. Values close to 1 indicate that the wind profiles at those locations are highly similar, exhibiting synchronized patterns, while values near 0 indicate no similarity. Each location can be conceptualized as a vector, either a row or a column of this symmetric matrix, capturing its correlation with all other locations. These vectors serve as the basis for clustering algorithms. The matrix dimension corresponds to the discretization described in the section 3.2: the onshore Australian area is gridded at 0.25° resolution in both longitude and latitude, resulting in 11,822 rows/columns.

3.7.1 Hierarchical clustering

The first clustering algorithm we used is hierarchical clustering, which organizes data points into a nested hierarchy of clusters by iteratively merging the closest pairs of clusters. Each data point corresponds to a location in the ERA5 reanalysis grid and is represented by a vector extracted from the wind speed correlation matrix. To quantify dissimilarity between points, we use the Euclidean distance (L2 norm) between their correlation vectors. Given two points $\mathbf{x}_i$ and $\mathbf{x}_j$ represented by their correlation vectors, the distance is defined as:

$$d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^{m}(x_{ik} - x_{jk})^2}$$

where m is the dimensionality of the correlation vectors (i.e., the number of locations, 11,822).

The hierarchical clustering algorithm proceeds by merging clusters based on a *linkage criterion*, which defines how the distance between two clusters A and B is computed from the pairwise distances between their members. Different linkage methods lead to different clustering structures, as described below:

- **Complete linkage** (farthest point):

$$d(A,B) = \max_{a\in A, b\in B} d(a,b)$$

- **Average linkage** (UPGMA):

$$d(A,B) = \frac{1}{|A||B|}\sum_{a\in A}\sum_{b\in B} d(a,b)$$

- **Weighted linkage** (WPGMA):

$$d(A,B) = \frac{d(A',B) + d(A'',B)}{2}$$

  where A' and A'' are the clusters merged to form A.

- **Centroid linkage** (UPGMC):

$$d(A,B) = \|\bar{\mathbf{x}}_A - \bar{\mathbf{x}}_B\|_2$$

  The distance is the Euclidean distance between the centroids (means) of the two clusters.

- **Median linkage** (WPGMC): Similar to centroid linkage, but uses median vectors instead of means.

- **Ward's linkage**: This method merges clusters using the Ward variance minimization algorithm

The hierarchical clustering was performed with a fixed number of 15 clusters, by cutting the dendrogram at the desired level (Fig. 3.11).

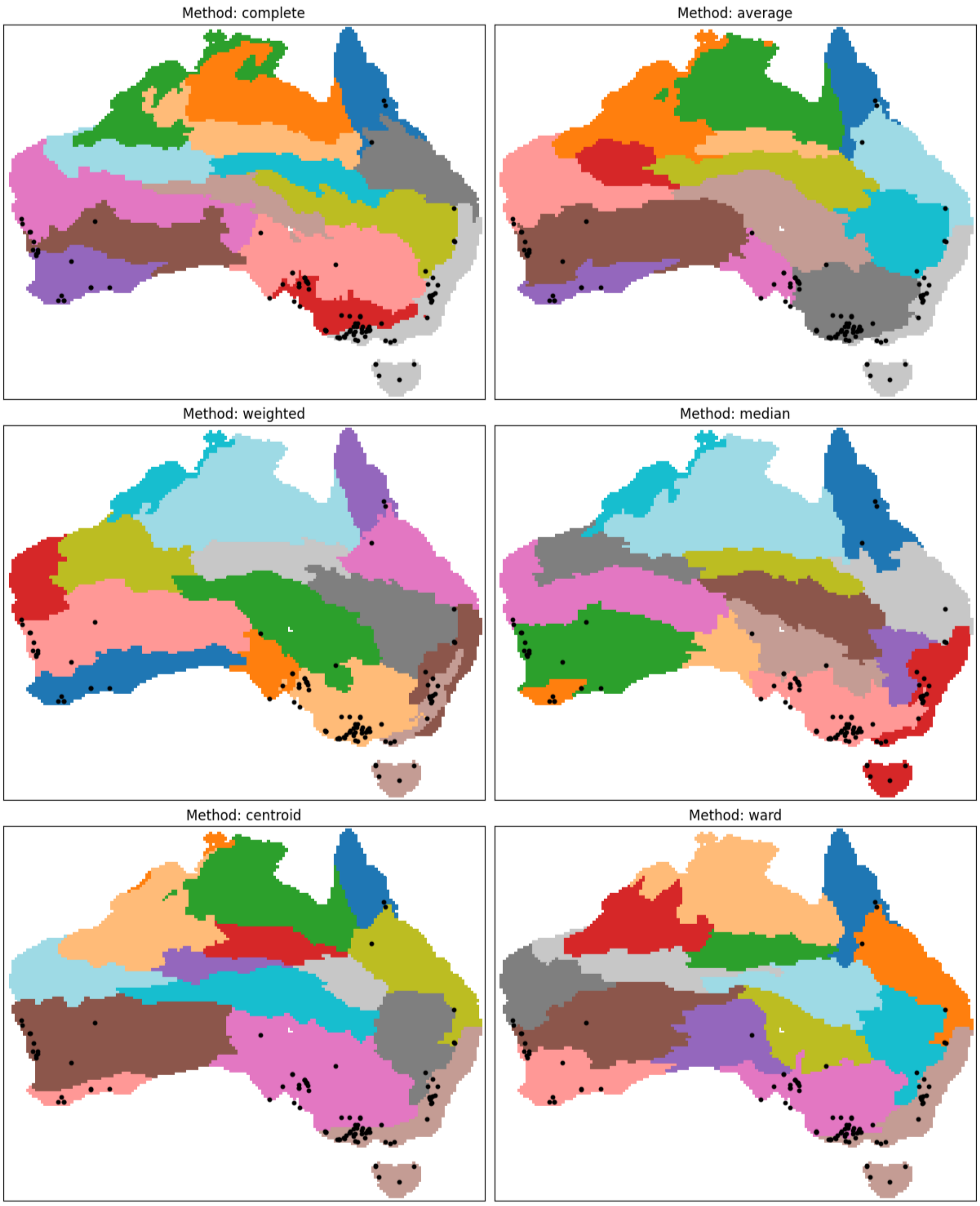


Figure 3.11: Hierarchical clustering results of wind correlation data using various linkage methods, the colored regions represent 15 distinct clusters (wind zones), while black dots indicate the positions of currently operating wind farms.

Linkage Method	Top 3 Clusters Coverage (%)
Centroid	85.4
Median	81.6
Ward	81.6
Complete	81.6
Weighted	77.7
Average	75.7

Table 3.1: The table shows the cumulative percentage of operating wind farms contained within the top three most populated clusters, meaning the clusters with the highest number of wind farms, for each hierarchical clustering linkage method.

3.7.2 K-means clustering

K-means clustering is a widely used partitional clustering algorithm that aims to partition the dataset into K clusters by minimizing the within-cluster sum of squares (WCSS). Given a set of data points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, the algorithm assigns each point to the cluster with the nearest centroid $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_K\}$.

Formally, K-means attempts to solve the optimization problem:

$$\arg\min_{\{\mathcal{C}_k\}} \sum_{k=1}^{K} \sum_{\mathbf{x}_i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2$$

where $\mathcal{C}_k$ denotes the set of points assigned to cluster k and $\boldsymbol{\mu}_k$ is the centroid of cluster k, computed as:

$$\boldsymbol{\mu}_k = \frac{1}{|\mathcal{C}_k|} \sum_{\mathbf{x}_i \in \mathcal{C}_k} \mathbf{x}_i$$

The algorithm proceeds iteratively with two main steps:

1. Assignment step: Assign each point to the nearest centroid based on Euclidean distance.
2. Update step: Recalculate centroids as the mean of all points assigned to each cluster.

In our analysis, the K-means algorithm is applied to the wind speed correlation vectors. We adopt the default implementation provided by `scikit-learn`. In particular, the algorithm uses the k-means++ initialization method, and convergence is declared when the change in inertia falls below a tolerance, or after a maximum of 300 iterations.

We set the number of clusters to 15 to maintain consistency with the hierarchical clustering experiments and allow for direct comparison.

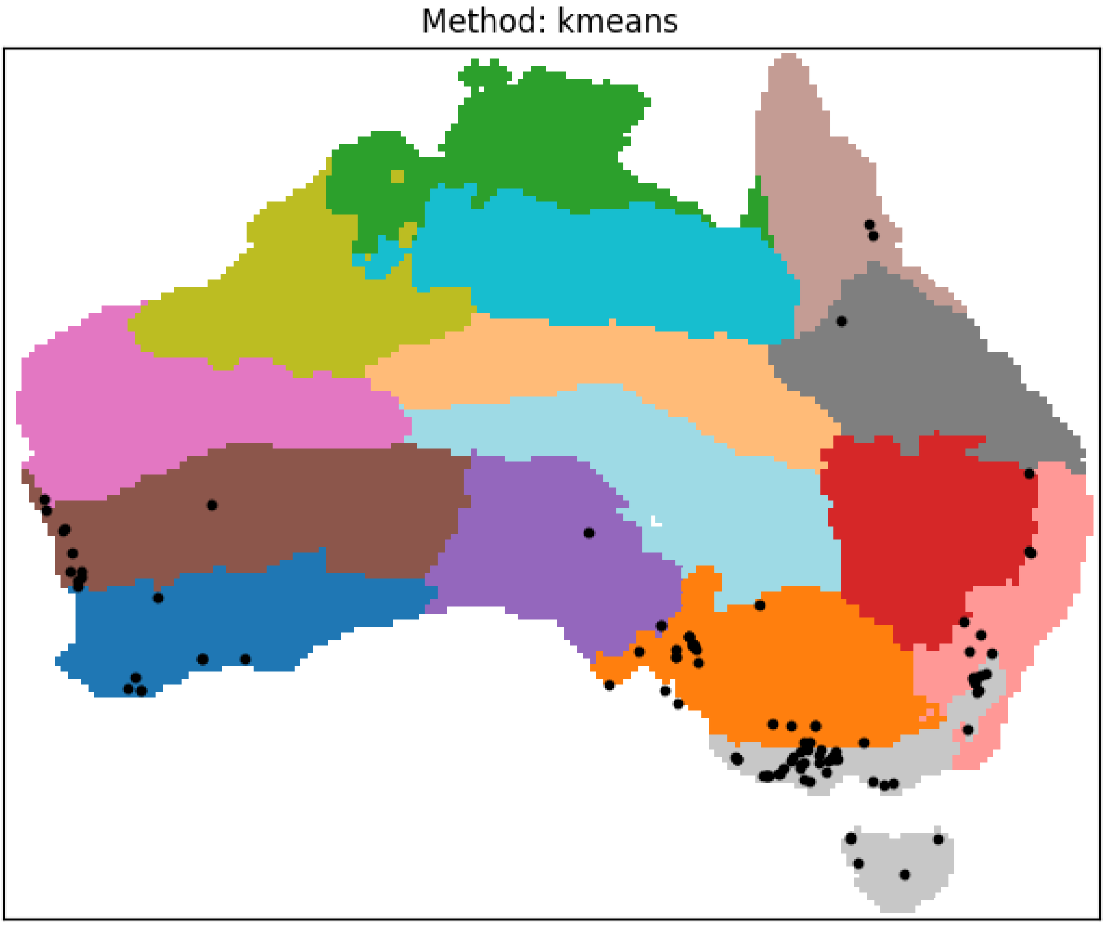


Figure 3.12: K-means with 15 clusters. Each color represents a distinct cluster. Black dots indicate the locations of operating wind farms.

Method	**Top 3 Clusters Coverage (%)**
K-means	75.7

Table 3.2: The cumulative percentage of operating wind farms contained within the top three most populated clusters produced by K-means.

3.7.3 Spectral clustering

Spectral clustering is a technique that uses the spectral (eigenvalue) properties of a similarity graph to identify clusters of data points that are connected or similar according to a defined notion of neighborhood. Unlike K-means or hierarchical methods that rely on direct distance measures in feature space, spectral clustering first builds a graph structure and then uses linear algebra to find an optimal low-dimensional representation in which clustering is easier. Once again we search for 15 clusters to enable direct comparison with other methods.

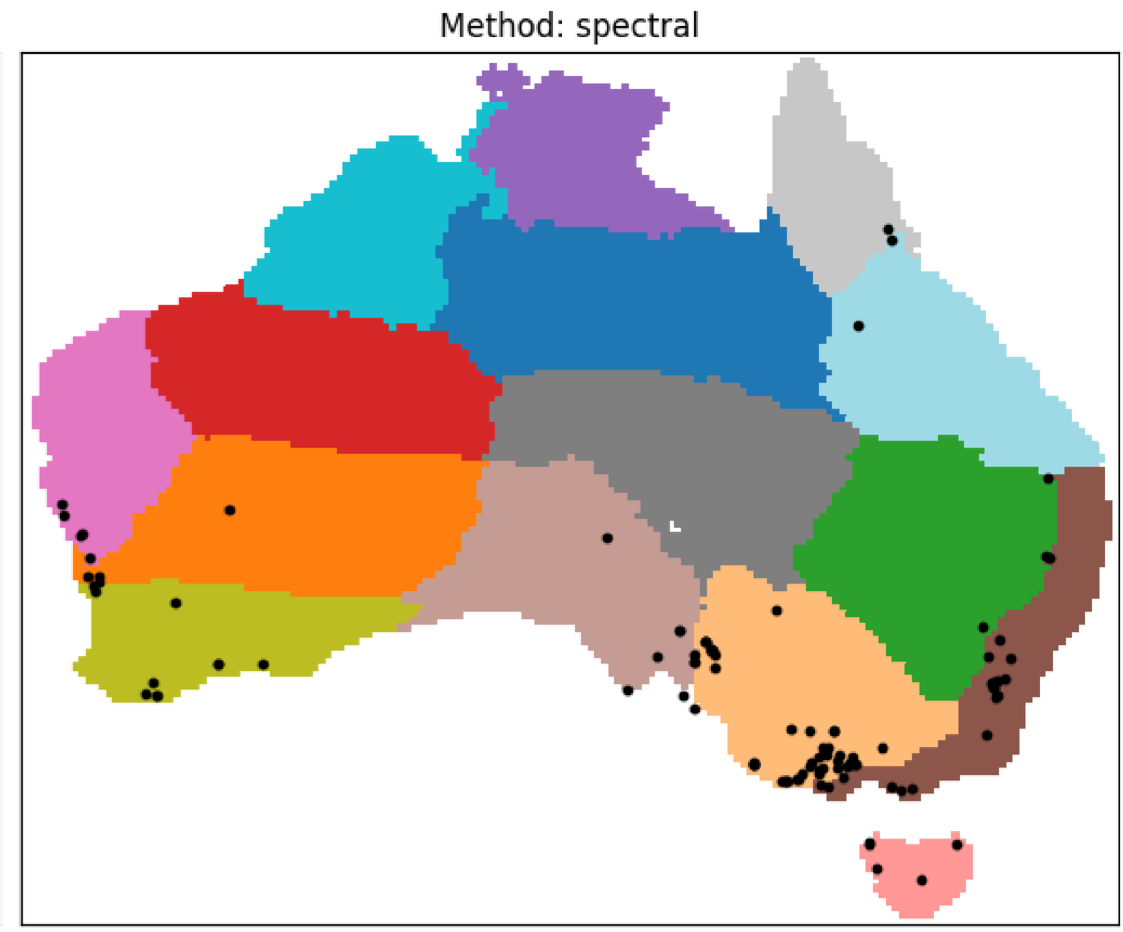


Figure 3.13: Spectral clustering into 15 clusters, using a nearest-neighbors affinity graph. Each color represents a different cluster. Black dots indicate the locations of operating wind farms

Method	Top 3 Clusters Coverage (%)
Spectral	71.8

Table 3.3: The cumulative percentage of operating wind farms contained within the top three most populated clusters produced by spectral clustering.

3.7.4 Gaussian Mixture Model clustering

As the final clustering method for comparison, we applied the Gaussian Mixture Model (GMM), a probabilistic clustering technique. GMM models the data as a mixture of multiple multivariate Gaussian distributions and assigns to each point the probability of belonging to each cluster.

Due to the high dimensionality of the correlation matrix, running GMM directly was computationally very expensive. To address this, we first applied Principal Component

Analysis (PCA) to reduce the dimensionality of the data, retaining the first 50 principal components. This reduction made the computation feasible while preserving most of the data's variance. We then performed clustering using a fixed number of clusters (15) to allow direct comparison with the other methods.

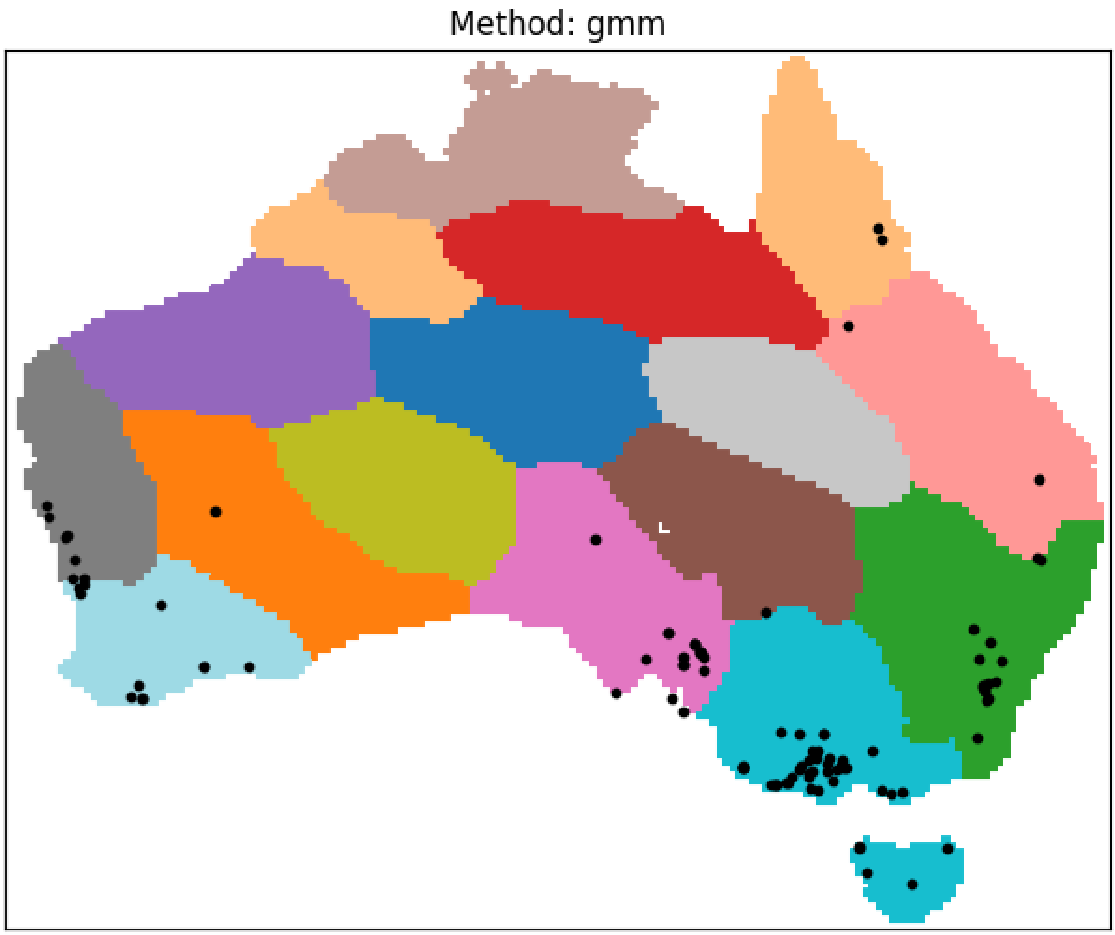


Figure 3.14: GMM clustering. Each color represents a different cluster. Black dots indicate the locations of operating wind farms

Method	Top 3 Clusters Coverage (%)
GMM	74.8

Table 3.4: The cumulative percentage of operating wind farms contained within the top three most populated clusters produced by GMM clustering.

3.7.5 Clustering conclusions

To conclude the clustering analysis, we observe that, regardless of the specific clustering technique employed, operating wind farms consistently appear highly concentrated within a small subset of wind clusters. When fixing the number of clusters to 15, the top 3 most populated clusters, which represent 20% of the total, contain between approximately 72% and 82% of all wind farms, as shown in the summary table 3.5. As a further example, when increasing the number of clusters to 50 (Fig. 3.16) and analyzing the cumulative share of wind farms in the top 20% of clusters (i.e., the 10 most populated ones), we observed even higher concentration levels, often exceeding 90%. This reinforces the observation.

The decision to fix the number of clusters at 15 was made to ensure clarity in visual interpretation. Identifying a single "correct" number of clusters is not particularly meaningful in this context, as our goal is not to uncover inherent cluster boundaries but to highlight spatial grouping tendencies. We also experimented with the elbow method applied to K-means (see Figure 3.15), but the curve did not display a clear inflection point.

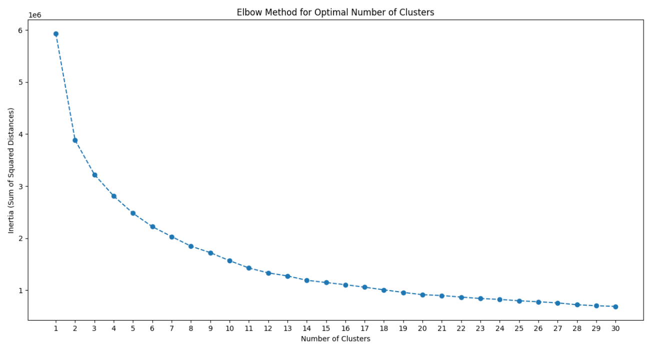


Figure 3.15: Elbow method applied to K-means clustering. There is no obvious 'optimal' number of clusters in this context.

Clustering Method	**Top 3 of 15 Clusters (%)**	**Top 10 of 50 Clusters (%)**
Hierarchical - Complete	81.6	91.3
Hierarchical - Median	81.6	87.4
Hierarchical - Average	75.7	93.2
Hierarchical - Ward	81.6	88.3
Hierarchical - Centroid	85.4	87.4
Hierarchical - Weighted	77.7	93.2
K-means	75.7	86.4
Spectral Clustering	71.8	87.4
Gaussian Mixture Model	74.8	93.2

Table 3.5: Cumulative percentage of operating wind farms contained within the most populated clusters for each clustering method. The first column corresponds to the top 3 clusters out of 15 (i.e., 20% of clusters), and the second column refers to the top 10 clusters out of 50 (also 20%).

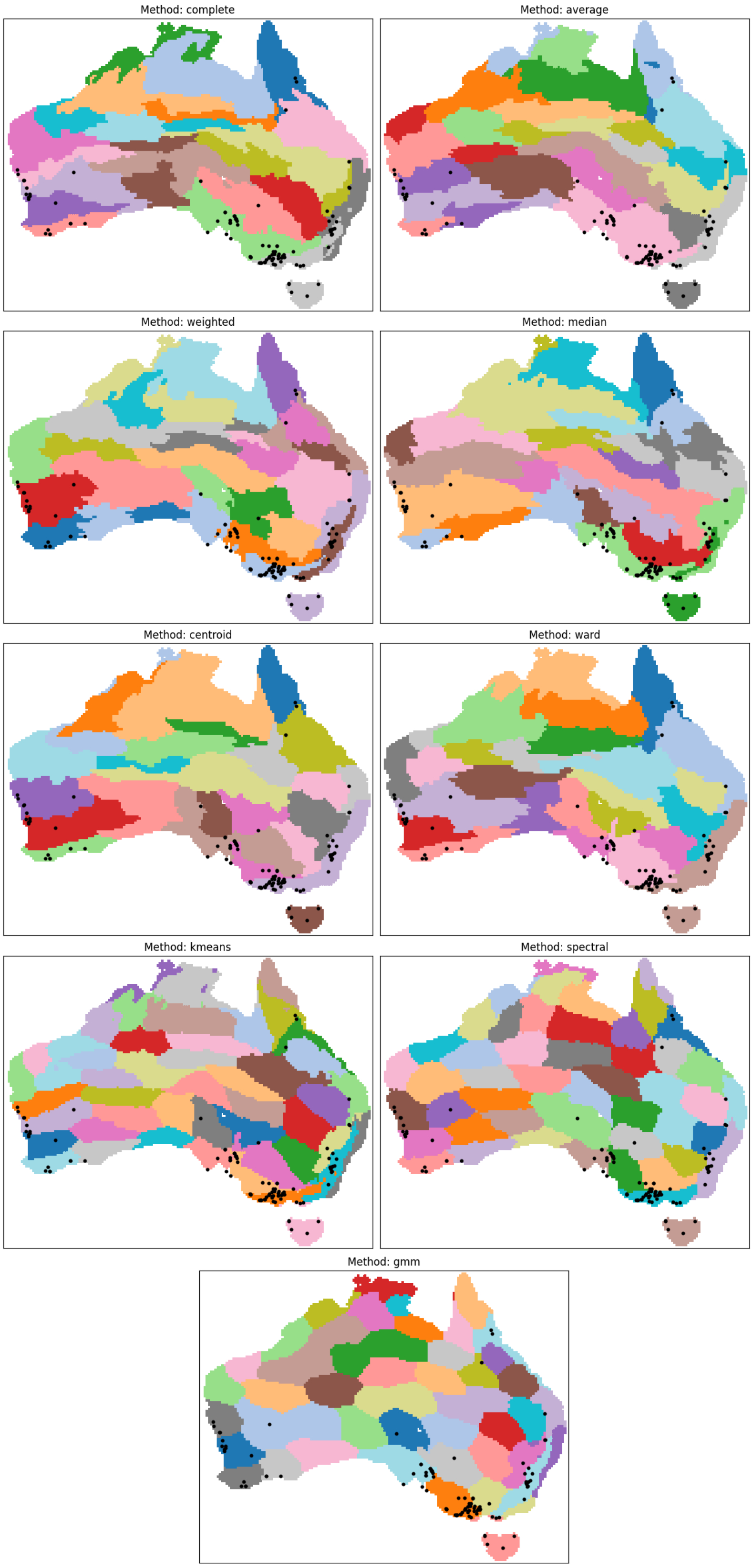


Figure 3.16: Clustering results using 50 clusters. Top 20% clusters account for around 90% of operating wind farms.

One particularly noteworthy aspect is that all clustering methods were applied solely to the wind speed correlation matrix (see Figure 3.10), without incorporating any explicit spatial coordinates or geographic information. Despite this, the resulting clusters consistently display strong spatial contiguity. This outcome is intuitively consistent with the nature of wind, which is inherently shaped by geographic factors, even when these are not directly included in the data.

In addition to the static results discussed above, the developed tool includes a dedicated interactive page for exploring clustering configurations in greater detail. Users can adjust the number of clusters and visually assess how wind behavior segments across the Australia based on the discussed methods. Furthermore, the interface supports similarity-based color mapping: by clicking on a cluster, the tool highlights others according to their average wind correlation similarity, enabling a more intuitive understanding of regional wind dynamics and potential relationships between areas (Fig. 3.17).

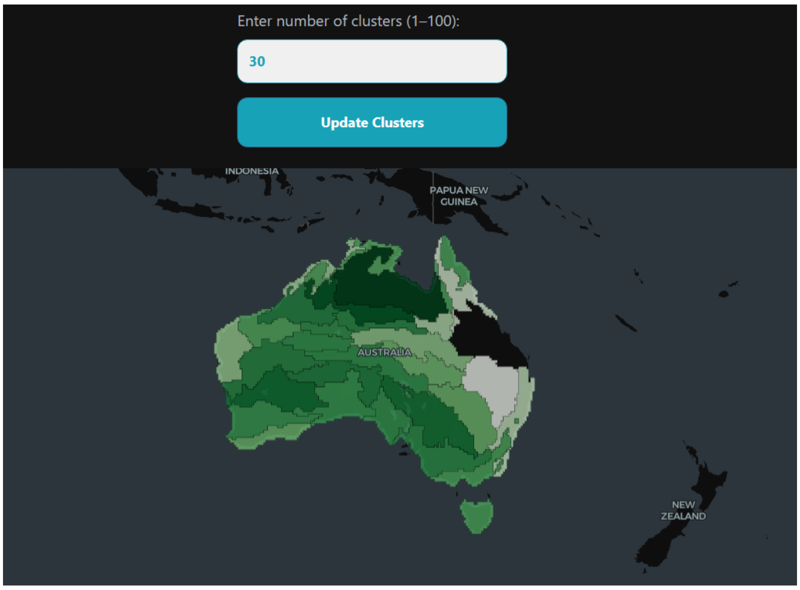


Figure 3.17: Users can select the number of clusters and click on a region (highlighted in black, the one in Queensland in the figure) to highlight others with similar or different wind profiles. Greener colors indicate higher inter-cluster distance. An example use case is selecting finer clusters within a specific state to identify a location for a new wind farm in a zone that is as uncorrelated as possible with existing infrastructure.

3.8 Explainable AI

While the primary focus of this work lies in the design and implementation of the Suitability Index and its interactive logic, the tool also includes additional pages and visualizations developed during the broader exploration of the wind farm siting problem. These components, though not central to the main tool, offer useful context and were instrumental in shaping the overall methodology. For brevity, only one is summarized here.

A side page included in the tool is the Local Explainers interface (Fig. 3.18). This section explores how explainable artificial intelligence (XAI) techniques can assist decision making providing localized insights into model predictions.

The interactive map displays which areas are deemed suitable or not suitable for wind development **by a machine learning model**. Unlike the Suitability Index page (see 3.4 and 3.5) , where users explicitly set the importance of each variable via weight sliders, this AI-based page follows a supervised learning approach: the model internally learns the weights of variables from labeled data.

By clicking on any location, users can explore the specific reasoning behind the model's classification, visualized through LIME (Local Interpretable Model-agnostic Explanations) [39] and SHAP (SHapley Additive exPlanations) [30], two widely used XAI methods for interpreting individual predictions.

To simplify the design and focus on explainability rather than pure prediction accuracy (more on it later), the training data was synthetically generated. The dataset is built using the suitability index logic computed with fixed weights. A random noise (with a mean offset of 10%) is added to each variable to simulate variability. Locations in the top 20% of the resulting scores are labeled as suitable (positive class), while the remaining 80% are labeled as not suitable (negative class). A Random Forest classifier is then trained on this synthetic dataset. Note: the goal of this module is not to produce a good siting model, but rather to demonstrate how local model interpretability could potentially complement siting models, offering more transparency into how and why a location is deemed suitable.

Problems of training a supervised model

The dataset that could be used for training a proper supervised model is extremely unbalanced, with approximately 12,000 possible locations but only between 100 and 200 labeled points, depending on whether we consider existing wind farms or also planned ones. Moreover, the main critical issue with supervised learning in this context is the model bias. When current wind farm locations are used as ground truth for training, the model essentially learns to replicate past bottom-up site selection decisions. In other words, the model risks automating a suboptimal decision-making process instead of improving upon it. That is why this setup is intended as a proof of concept and the used dataset is synthetic, demonstrating how XAI methods can help users understand and trust model based site selection if a more advanced predictive model is used in the future.

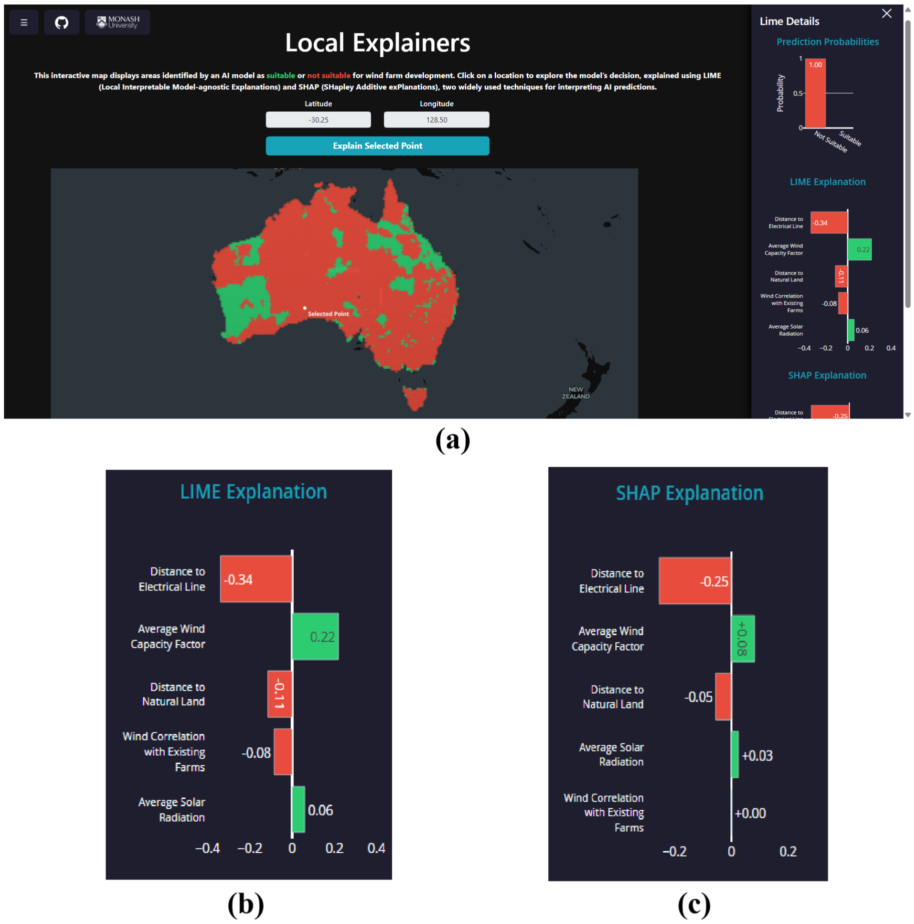


Figure 3.18: The *Local Explainers* page displays a map (**a**) of predicted suitable (green) and non-suitable (red) locations for wind farm development, as classified by a Random Forest model trained on synthetically generated data. Users can click on individual points to open a side panel and view local explanations of the model's decision using LIME (**b**) and SHAP (**c**), which reveal the feature importances influencing the prediction for the selected location. In this example, both explainers agree that the clicked location was classified as 'non-suitable' primarily due to its distance from the electrical line.

Conclusion and Future Work

Selecting sites for wind farms is not an easy task. It involves numerous variables, from environmental and social factors to economic and technical considerations. Moreover, this decision impacts the public and the surrounding communities in various ways, such as through land use, aesthetics, and even local economies. Given the complexity of these factors, several different methods have been developed to assist in the decision making process. Regardless of the method used, most of these approaches ultimately result in static visualization maps generated through GIS systems. This document suggests that there is a gap in the understandability and communication of how these maps are created, and in general, on how the reasoning behind the whole site selection is explained. To address this, I propose the implementation of a visual interface designed to help mitigate this issue by providing a more intuitive and accessible way to visualize and understand the underlying processes behind the creation of a suitability map. This interface could assist in making the process clearer for all parties involved. The static nature of conventional maps limits users' ability to interact with the data, create "what-if" scenarios, or explore the trade-offs inherent in site selection decisions. Often, these maps present a single best solution or a fixed ranking without revealing how different variables influence the overall suitability score or how prioritizing one factor over another might change the outcome. This work presents a novel interactive approach to the problem. The developed tool incorporates a suitability index based on weights defined by users, as well as Pareto optimization, a powerful method from multi-criteria decision analysis that identifies sets of optimal trade-off solutions. It also introduces a new variable that describes wind correlation with existing farms, which has never before been integrated into such suitability analyses. Additionally, results from the backtesting of this suitability index demonstrate how even a simple top-down approach that includes this new variable outperforms traditional bottom-up planning.

This work was also part of a broader interdisciplinary research effort involving a series of workshops aimed at exploring the role of interactive visualisation in the context of geospatial data. The workshops followed a mix of methods based on the Creative Visualisation Opportunities (CVO) framework. A total of 23 different academic researchers from Monash University (7 from the same research team) participated. The experts were from various domains such as power engineering, network optimisation, environmental AI, and climate science. All participants had varying degrees of knowledge in visualisation, AI, or geospatial data. They were recruited via contacts based on their expertise and interest.

One of the aims of the final workshop was to gather structured feedback for improving the tool. The workshop was designed around three activities. First, participants were asked to record their initial thoughts and aspirations regarding how AI and geospatial data could be applied to renewable energy, or other domains, prior to seeing the tool. The second activity involved a walkthrough of the ideas behind the prototype's design, followed by its live demonstration. During this part several use cases where shown, participants were encouraged to take notes (see Fig 4.3) and ask questions in real time. Finally, participants were split into two groups: one focused on the technical aspects of the developed tool, and the other on the interface and user experience. Each group was asked to reflect on and discuss potential improvements, as well as current and future use cases for the tool. In addition to the formal workshops, another valuable feedback opportunity came through a lab presentation held as part of the Data Visualization course at Monash University. During this session, I was invited to present the tool to a group of approximately 20 students enrolled in the course. The presentation included a brief overview of the project followed by a live demonstration of the tool's features. Students were then given the chance to interact directly with the interface and test various use cases.

The feedback derived from these sessions directly informed the design of the tool. In particular, some modifications and improvements were made, including:

- **Inclusion of a documentation page**: during discussions with users, it became clear that a better approach was to separate the explanations of variables and methodological details from the main interface. As a result, a dedicated documentation page was added, allowing the Suitability Index page to remain cleaner and more minimal while still ensuring that detailed information was accessible when needed. (See Figure 4.1)

- **Interactive masking of energy infrastructure**: to allow users to interactively include or exclude specific layers of energy infrastructure (such as the electrical grid, see Figure 4.2) from the suitability analysis. This would help simulate planning scenarios where certain constraints or developments are assumed to be added, removed, or ignored.

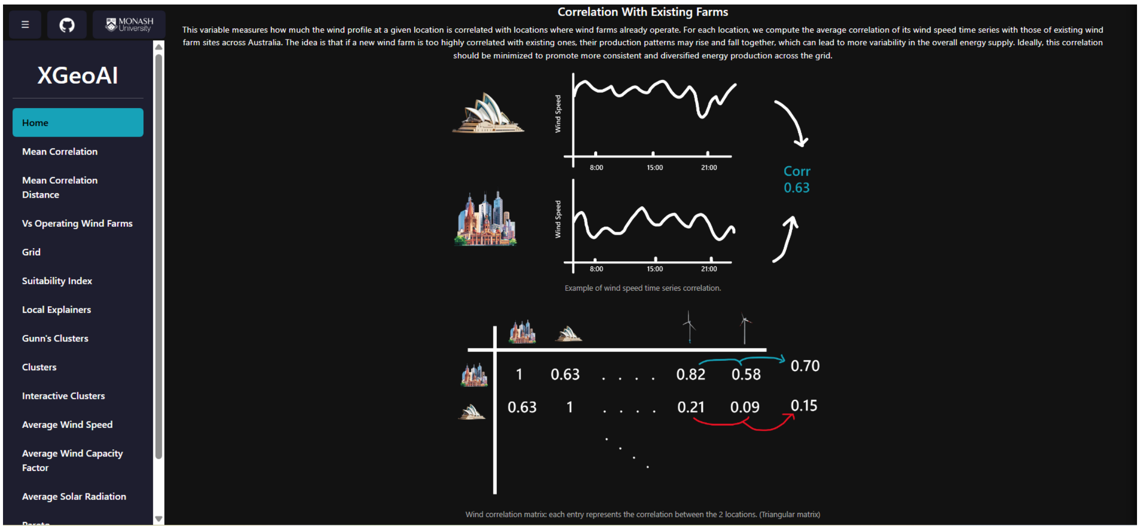


Figure 4.1: Screenshot from the documentation page, showing the explanation for one of the five variables used in the Suitability Index. This dedicated page was added based on user feedback to keep the main interface minimal while still offering detailed, accessible information about each input parameter.

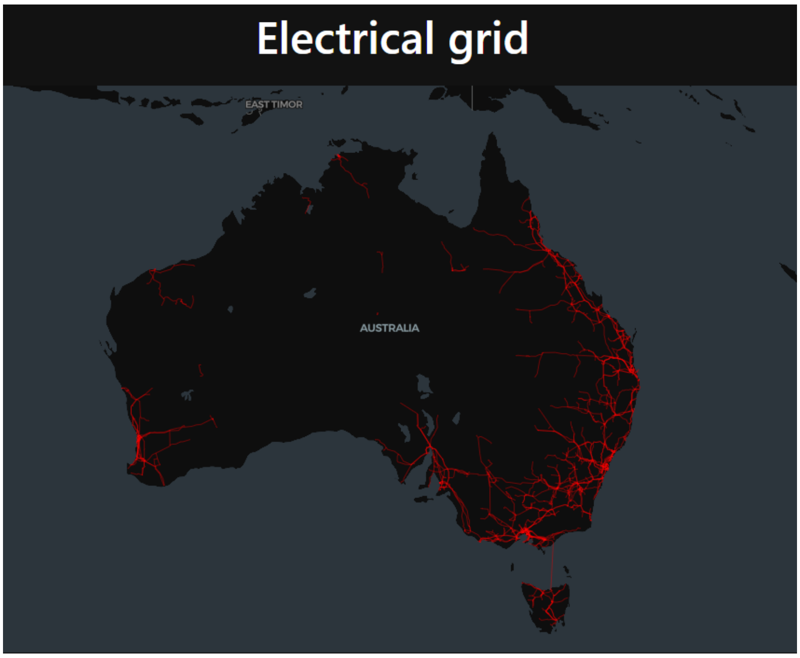
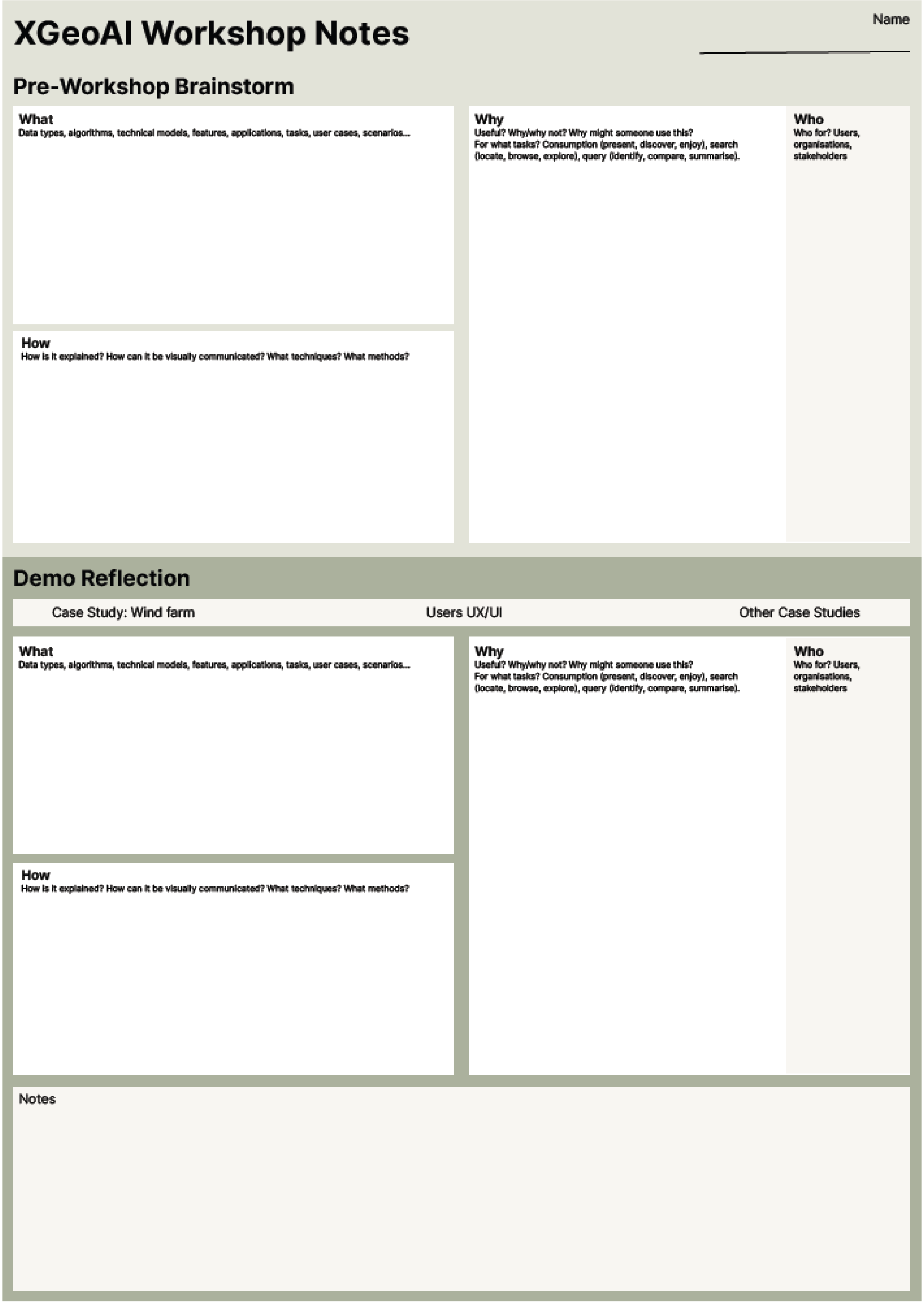


Figure 4.2: Overlay of the electrical grid on the map. Included as a result of feedback sessions.

Another proposed improvement was the addition of a time selection feature, enabling users to explore how wind farm suitability varies across different years and better account for temporal variability in weather patterns. While users appreciated the ability to adjust input weights, there was a clear demand for more guidance in assigning these weights, including predefined “expert weight profiles” accompanied by information about the contributing experts to increase transparency.



XGeoAI Workshop Notes

Name

Pre-Workshop Brainstorm

What
Data types, algorithms, technical models, features, applications, tasks, user cases, scenarios...

Why
Useful? Why/why not? Why might someone use this?
For what tasks? Consumption (present, discover, enjoy), search (locate, browse, explore), query (identify, compare, summarise).

Who
Who for? Users, organisations, stakeholders

How
How is it explained? How can it be visually communicated? What techniques? What methods?

Demo Reflection

Case Study: Wind farm | Users UX/UI | Other Case Studies

What
Data types, algorithms, technical models, features, applications, tasks, user cases, scenarios...

Why
Useful? Why/why not? Why might someone use this?
For what tasks? Consumption (present, discover, enjoy), search (locate, browse, explore), query (identify, compare, summarise).

Who
Who for? Users, organisations, stakeholders

How
How is it explained? How can it be visually communicated? What techniques? What methods?

Notes

Figure 4.3: Form provided to participants during the final workshop for structured feedback. The form was designed to capture participants' initial expectations before the demo, notes during the live walkthrough, and reflections or suggestions after seeing the tool.

Many of the suggestions gathered from user feedback naturally fall within the broader scope of future work for this project. While the current tool demonstrates the feasibility and advantages of an interactive approach to wind farm site selection, it remains a proof of concept rather than a fully comprehensive solution. At present, the model incorporates only five variables, which limits its scope and detail. However the framework is designed to be easily extendable to include the full range of relevant variables for real applications. Future development could also focus on integrating specific domain knowledge to refine and improve the suitability index, moving beyond the current linear combination which may overlook complex interactions. While the tool is explainable by default and machine learning techniques, along with the associated XAI methods, have been incorporated as supplementary tools, future iterations could explore more diverse modeling approaches that go beyond the limitations of supervised learning in this context. Additionally, the current analysis relies on 11,822 data points; therefore, if users zoom into specific regions, this resolution may be insufficient. To address this, more detailed data sources than ERA5 might be necessary, or sophisticated interpolation techniques could be employed. Although interpolation of wind data is a complex field in its own right.

Finally, this work has focused exclusively on onshore wind turbines; however, offshore wind farms present their own set of challenges that could also benefit from a similar interactive and explainable decision support approach in future extensions of the tool.

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